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A PROJECT REPORT ON
**“Process Development for production of an Active Pharmaceutical
Ingredient (API) Precursor”**

SUBMITTED TO
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2024



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CERTIFICATE

THIS IS TO CERTIFY THAT THE PROJECT REPORT TITLED

**“Process Development for production of an Active Pharmaceutical Ingredient (API)
Precursor”**

SUBMITTED BY :

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In partial fulfilment of requirement of Capstone Project, is a bonafide record of the work carried out by him during the period of his research internship from January 2024 to May 2024. He has fulfilled the requirement of the submission of the Capstone project of the Final Year Chemical Engineering as per the syllabus prescribed by MIT World Peace University, Pune. The material from other sources has been duly acknowledged in the report.

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CAPSTONE PROJECT REPORT TITLED

**“Process Development for production of an Active Pharmaceutical Ingredient (API)
Precursor”**

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DECLARATION

I hereby declare that this submission is representing my ideas in my words. The ideas and words included from other sources are adequately cited. I also declare that I have followed all principles of academic honesty and integrity of the University. I have not misrepresented or fabricated or falsified any fact/data/idea in this submission. I understand that any violation will cause disciplinary action against me by the University. I also understand that there will be a penal action against me, from the sources, which have thus not been properly cited or from whom proper permission has not been taken wherever needed.

Abstract

Sustainable process for production of Active Pharmaceutical Ingredients (API) precursor has recently gained a lot of attention due to changes in manufacturing pattern of API after COVID-19 pandemic. Through this study, a new process for production of an important API precursor “C” has been developed. This novel process replaces the liquid acid catalyst used in the conventional process with the solid acid catalysts. The solid acid catalysts were studied using ^{31}P Nuclear Magnetic Resonance (NMR) Spectroscopy. For obtaining best results, solid acid catalysts that fitted the required parameter identified from characterization were screened and the ones giving best results were selected. Several exploratory experiments studying reactant-product interaction with catalyst, oligomerization and side reactions and deactivation of catalysts were carried out. The spent catalyst was characterized using ^{31}P NMR Spectroscopy. The products obtained after reactions were analysed using Gas Chromatography, High Resolution Liquid Chromatography Mass Spectrometry. For separation and isolation of “C” distillation was carried out. A separate apparatus was thought of and designed for the regeneration of catalyst.

Keywords – API Precursor, Solid acid catalysts, ^{31}P NMR, Process development

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1) Introduction

Active Pharmaceutical Ingredients (APIs) serve as the fundamental constituents of pharmaceutical products, delivering their therapeutic effects. Factors such as high production costs and environmental sustainability are taken with more awareness and the recent COVID-19 pandemic have produced a change in the manufacture of APIs, promoting their importation from foreign countries. To return to local manufacturing, numerous state agencies are developing various initiatives that promote the production of APIs by following the principles of Green Chemistry with the aim of not harming the environment and complying with environmental regulations. The raw materials which form the base and are used in manufacturing of API are known as API Precursors. The selection and synthesis of these API Precursors play a very important role in ensuring the quality, safety and effectiveness of the drug product. Thus, for the green synthesis and large scale manufacturing of API, a sustainable process for manufacturing the API Precursor needs to be developed.

This primary experimental study is aimed at developing a sustainable, cheaper and highly efficient process for manufacturing an industrially important API Precursor “C”. Conventionally, the process of synthesis of C involves reaction of A derived from refining of petroleum product with acylating agent B using a liquid acid catalyst. The process which we aim to develop and deploy in the industry focusses on the reaction of A obtained from Bio-refinery with acylating agent B using a solid acid catalyst. The use of A obtained from Biorefinery and solid acid catalysts makes the process cost competitive while also making it sustainable.

The crucial step for the development of the process is the screening and selection of solid acid catalyst which gives the best performance for our reaction. Instead of screening all the available catalysts and relying on the trial and error method, we have characterized and identified the features and characteristics of solid acid catalysts which are responsible for specific catalytic activity. These characteristic features include the type of acid sites, number of acid sites, strength of acid sites and the distribution of acid sites. The catalysts which match the specific features required for high conversion of A and high yield of C were screened and the best of the catalysts were selected. The reactions were carried out using these selected solid acid catalysts in batch mode of operation as well as continuous mode of operation.

Oligomerization reactions occur simultaneously with the main reaction. As a result, large chain oligomers are formed which are responsible for deactivation of the catalyst. Set of experiments were conducted by us in order to track these oligomerization reaction products and to study the deactivation of catalysts. Sophisticated analytical and characterization techniques like Gas Chromatography (GC), High Resolution Liquid Chromatography Mass

Spectroscopy (HRLCMS) and ^{31}P NMR were used for analysis of the products formed and study of the solid acid catalyst.

In order to separate the product of interest from the product mixture, a distillation apparatus was set up. Several distillation runs were carried out to perfect the standard operating procedure and the general methodology along with the process conditions for carrying out the distillation operation. The product obtained after distillation was isolated and double distilled once again to ensure the high purity of target product C.

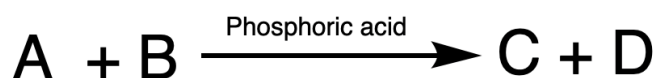
Regeneration of the spent catalyst was another industrially important factor which played a very important role in increasing the profitability and efficiency of the process. A new design for regeneration of catalyst utilizing fluidized bed was thought of and modelled. By taking into account the catalyst size, porosity and regeneration kinetics; minimum and maximum fluidization velocities were calculated. A theoretical design for apparatus for regeneration of catalyst was developed.

A systematic approach was followed while doing literature review, designing the experiments and interpreting the results of the experiments. Basic principles of experimental design like principle of replication, principle of randomization and principle of local control were taken into account while designing the experiments. Inferential statistics was used extensively to estimate the characteristics of the sample group while accounting for errors and also to test the research hypothesis of a given sample. As a result of use of this systematic approach and the robust techniques, the study on development of process for production of API Precursor was performed successfully and conclusive promising results were obtained.

2) Literature Review

2.1] Conventional Process

The current process used widely in the industry for the production of API Precursor C uses raw materials obtained from processing of crude oil. The process also utilizes liquid acid catalyst for catalysing the acylation reaction for producing the product C. The process is carried out in batch mode of operation. The liquid acid catalyst used for the reaction is Phosphoric acid. The reaction has been carried out using several other liquid acid catalysts of varying strength like Sulphuric Acid and Triflic Acid and the results have been noted to investigate the effect of acid strength on the conversion of reactants and yield of products. It has been observed that as the strength of the liquid acid is increased, the conversion of reactants increases greatly, however, the rate of oligomerization i.e. the side reactions also increases thereby decreasing the yield of the product C. For the acylation of interest, the acylating agent B acts as a Brønsted base. Therefore, Brønsted acid catalyst must be used for facilitating the acylation reaction by activation of B.



For obtaining high yields of C, the acid catalyst (phosphoric acid) and B are mixed evenly and stirred in order to ensure uniform heating and concentration of the mixture. The component A is then added dropwise to the mixture. The temperature is maintained at 45 °C during the complete process. Once the reactant B is exhausted, the reaction is stopped and the product mixture is cooled down to room temperature. The mixture is then extracted with solvent three times. The acid is then neutralised with an alkali and the pH of the solution is maintained at 7. The mixture is then transferred to a separating funnel where it is allowed to settle. The organic layer is separated from the aqueous layer and sent for distillation. Using vacuum distillation the pure product C is obtained.

The use of liquid acid catalyst for producing C has several disadvantages. Liquid acids catalysts being in the same liquid phase as that of the reactants and products, form a homogenous mixture. Thus, in order to separate the pure product, unreacted reactants and the catalyst, sophisticated distillation setup is required. During the course of the reaction, the liquid acid catalyst reacts with the side products formed. The liquid acid is also neutralised for efficient separation of the organic layer and aqueous layer. Thus, for every run fresh liquid acid catalysts need to be utilized. The process of separation of the liquid acid

catalyst from the product mixture using extraction, neutralization is also a tedious process for the industry. Moreover the capital investment required for this process is significantly higher thereby providing scope for improvement of the conventional process.

2.2] Use of solid acid catalyst

Liquid acid catalysts like HF, H₂SO₄ and H₃PO₄ are one of the first acid catalysts and are widely used in the petrochemical and chemical manufacturing industry. However, since 1940, the trend has changed and the liquid acid catalysts are being replaced by solid acid catalysts wherever possible(1). Solid Acid catalysts have several benefits as compared to the liquid acid catalysts. First and foremost, the separation of product and catalyst is easier as compared to liquid acid catalysts. Catalyst can be separated from the product mixture simply by the use of filtration. Some of the solid acid catalysts can also be regenerated and reused in the process thereby reducing the waste and making the process environmentally friendly.

Most commonly used solid acid catalysts are silico-aluminates (zeolites, silico-aluminophosphates), sulphated zirconia, transition metal oxides (Titania, Niobia), Niobic acid and Ion Exchange Resins (Polystyrene Sulfonates). It has been noted that for the acylation reaction of A, Brønsted acid sites need to be present. Through various experiments performed over the years, silico-aluminates (zeolites) and Ion Exchange Resins have been found to show the best performance.

Zeolites mainly consist of Silicon, Aluminium and Oxygen. They have the general formula (MAlO₂)(SiO₂)_x(H₂O)_y where M⁺ is usually H⁺ or Na⁺. Zeolites are mostly crystalline with primary building blocks of TO₄ with T being Si⁴⁺ or Al³⁺ cation at the centre surrounded by four oxygen atoms at the corners. These corner oxygen atoms are shared by two TO₄ units giving rise to a three dimensional structure with cavities and channels which are often referred to as “micropores”.

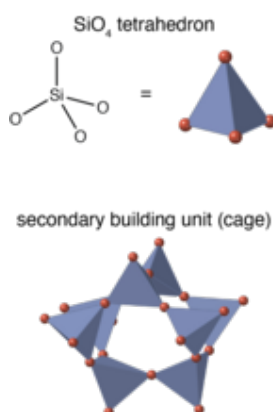


Figure 1- Structure of Zeolites

The ratio of Si/Al greatly affects the properties of solid acid catalysts. The difference in charge of Si^{4+} and Al^{3+} , causes the TO_4 unit to have a net neutral charge (SiO_4) or a net negative charge (AlO_4). Due to the presence of Al^{3+} in the framework, there is appearance of negative charges that requires the presence of charge balancing cations to ensure the electroneutrality of the solid acid catalysts. These charge balancing cations mostly occupy the micropore space and are not grafted into the zeolite framework. These charge balancing species are bounded to the lattice by Coulombic forces (2). Therefore, these charge balancing species can be partially or completely exchanged with different cations. This property is put to use when Zeolites are used in detergent formulations as a water softener. The Na^+ ions which are loosely bound to the framework are replaced by Ca^{2+} ions present in the hard water (3). As the Si/Al ratio of the zeolites is altered the presence of the charge balancing species increases or decreases. The presence of the charge balancing species dictates the acidic characteristics of the solid acid catalyst. The Si/Al ratio is responsible for generation of Brønsted and Lewis acid sites in Zeolites. In purely siliceous zeolite framework, the surface has no acidity on its own, however, the acid sites are generated due to structural defects. Development of Brønsted acid sites are related with Isomorphous substitution of Si^{4+} with Al^{3+} . This is because, substitution of Si^{4+} with Al^{3+} generates negative charge on the framework which is then compensated by the presence of protons. The Brønsted acid sites usually arise at oxygen atoms of a hydroxyl group bonded to a metal atom of one type coordinating to an atom of another type, which acts as a Brønsted base – the bridging hydroxyls (4).

Ion exchange resins are cross linked, water insoluble, ionisable functional groups. The ionisable groups are introduced to the polymer backbone after the process of polymerization and are distributed evenly along the polymer backbone. Most commonly used resins are cross linked polystyrene and polymethacrylate polymers (5). The synthetic ion exchange resins are usually cast as porous beads with considerable external and pore surface which allows the ions to attach and facilitates ion exchange. The resins are prepared as spherical beads usually 0.5 to 1 mm diameter. The major advantage of Ion exchange resins over the zeolites is that the surface area and the number of available active sites of the Ion exchange resins is much higher as compared to that of the zeolites (6).

The ion exchange resins are broadly classified into 4 main groups :

- 1) Strong Cation Exchange – containing sulfonic acid groups or corresponding salts
- 2) Weak Cation Exchange - containing carboxylic acid groups or corresponding salts
- 3) Strong Anion Exchange - containing quaternary ammonium groups
- 4) Weak Anion Exchange – containing ammonium groups.

Out of these four groups of Ion exchange resins, for the reaction of our interest, strong cation exchange resins are favourable. The reason for this is the presence of highly acidic Brønsted sites.

The general structure of an strong cation exchange resins is shown in figure.

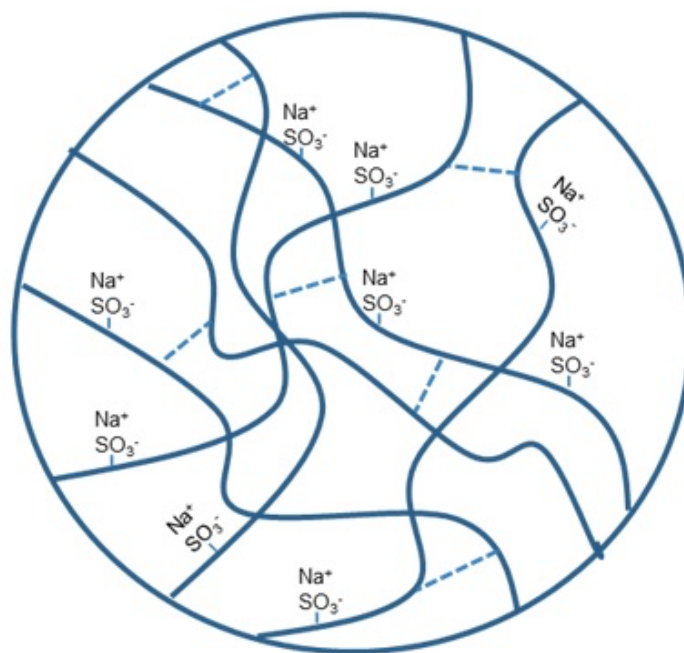


Figure 2 - Structure of Ion Exchange Resin

2.3] Improvement of conventional process

The advantages of solid acid catalysts have been well understood. Considering the environmental and economic advantages of solid acid catalysts over liquid acid catalysts, it is advisable to replace the liquid acid catalysts wherever possible. While replacing the liquid acid catalyst with solid acid catalysts, the conversion of reactants, selectivity of the target product and the yield of the target product should not be affected in a negative way. This study aims at improving the conventional process of production of industrially valuable product C by replacing the liquid acid catalyst with solid acid catalysts while also attempting to improve the conversion of reactant A and yield of product C. The use of solid acid catalysts comes with obvious benefits like ease in separation of catalyst from the product mixture and reuse of the catalysts by regeneration which decreases the capital investment.

The use of solid acid catalysts alters the reaction mechanism and also the rate of reaction. The acylation reaction of A over solid acid catalyst follow Eley-Rideal mechanism. The first step of the reaction is the adsorption of reactant B on the catalyst surface. Surface reaction of B with solid acid catalyst yields co-product D and highly reactive electrophile. As the co-

product D leaves the acidic site, reactant A attacks the acid site. Reactant A undergoes surface rearrangement reaction to transform into a strong nucleophile. The highly reactive electrophile formed from the surface reaction of B attacks the nucleophilic centre of reactant A and leads to formation of product C. Product C is desorbed in the last step of the mechanism. The attack of the highly reactive electrophile on the nucleophilic centre of A is the rate determining step for the mechanism.

For acylation of A Brønsted acid sites zeolites are particularly efficient and add great value and functionality to the bio-derived compounds. Solid Lewis acid zeolites are isomorphously substituted with tetravalent metals that act as isolated framework Lewis acid sites. The activity and selectivity of these framework Lewis acid sites is much higher as compared to the Brønsted acid sites for conversion of biomass like glucose isomerization, reduction and esterification. However, their activity is not yet investigated for acylation reactions. Thus, it is of great interest to determine how Lewis and Brønsted acid sites affect the acylation reaction of A. The strength of Lewis and Brønsted acid sites also greatly affects the rate of reaction, conversion of A and yield of C. Thus, in order to screen and select the perfect catalyst which can make the process efficient and provide maximum conversion of A and yield of C, we need to understand the characteristics of solid acids.

2.4] Characterization of Acidity in Solid Acid Catalysts

The predominant features of acidic zeolites which affect the activity of catalyst during the catalytic reactions are acid type (Brønsted Vs Lewis), strength, concentration and distribution(7). A comprehensive understanding of these individual acidic properties is very crucial for the screening, design, modification and utilization of solid acid catalysts in the industry. For specific uses, the specific acidic features can be tuned according to the need to get desirable results. For characterization, quantification and analysis of these individual acid properties, extensive spectroscopic and analytical studies have been carried out.

Infrared Spectroscopy (IR) – Mass Spectroscopy (MS) along with Temperature programmed Desorption (TPD) of ammonia has been used to study the nature, strength, crystallographic location and the distribution of acid sites in zeolites. In this technique, the IR-MS work together to follow the thermal behaviour of adsorbed and desorbed ammonia (8)(7). Fourier Transform Infrared spectroscopy (FTIR) has also been widely used to characterize the nature and the number of acid sites present in the zeolite (7,9,10). The catalytic cycle usually involves adsorption steps, surface reaction process and desorption process. The energetics of these surface chemical events play a very important role in determining the catalytic properties of the surface of the solid acid catalysts. The analysis of energetics is covered in the microcalorimetric studies which give a great insight in the features of solid acid catalysts(11,12). Calorimetric titration is another method which analyses the heat of adsorption distributions obtained by titration. Using this data, the surface acidity of the solid

acid catalysts and the distribution of acidic sites can be found out (13,14). Several Nuclear Magnetic Resonance (NMR) techniques have also been studied to investigate the nature and characteristics of the solid acid catalysts (7,15). However, the above conventional methods of studying the nature and features of solid acid catalysts utilize either the direct detection of Hydroxyl groups (OH) in solid acid catalysts by FTIR or ^1H Magic Angle Spin (MAS) NMR spectroscopy or through the adsorption of basic probe molecules like ammonia or pyridine. However, unlike direct detection of hydroxyls, most of the above indirect techniques invoke evaluation of heat of adsorption/desorption associated with the basic probe and acid sites, which may be correlated to the Brønsted acid sites. The qualitative determination of acid properties in solid catalysts often relies on similar principles, utilizing techniques like temperature programmed desorption (TPD) for ammonia and Fourier-transform infrared (FTIR) spectroscopy for pyridine and carbon monoxide. The commonly employed techniques, such as NH_3 -TPD and pyridine-IR, utilize probe molecules that are typically too alkaline to discern subtle variations in the distribution of acidic sites. As a result, they generate basic profiles that only indicate the overall strength of acidity on average (16). While certain traditional methods can swiftly measure Brønsted and Lewis acidities, offering insights into their quantity and potency, they often fall short in revealing the distribution and precise locations of acid sites.

2.5] NMR – Spectroscopy

Among the array of experimental approaches for characterizing acidity, NMR stands out as the most adaptable technique (7). In order to understand the application of NMR spectroscopy for acidity characterization, the basic working of NMR spectroscopy needs to be understood. Nuclear Magnetic Resonance (NMR) is a spectroscopic technique based on reorientation of atomic nuclei with non-zero nuclear spins in an external magnetic field. In order to obtain a NMR spectra, the NMR active sample is placed in a constant magnetic field B_0 . NMR-active samples refers to the samples whose constituent compounds and elements have a nucleus with a non-zero nuclear spin. The magnetic moment of the nucleus forces the nucleus to behave like a tiny magnet. In the absence of any external magnetic field, each tiny magnet (nucleus) is randomly oriented. The application of external magnetic field causes the nuclei to align with (low energy) or against (high energy) the external magnetic field B_0 .

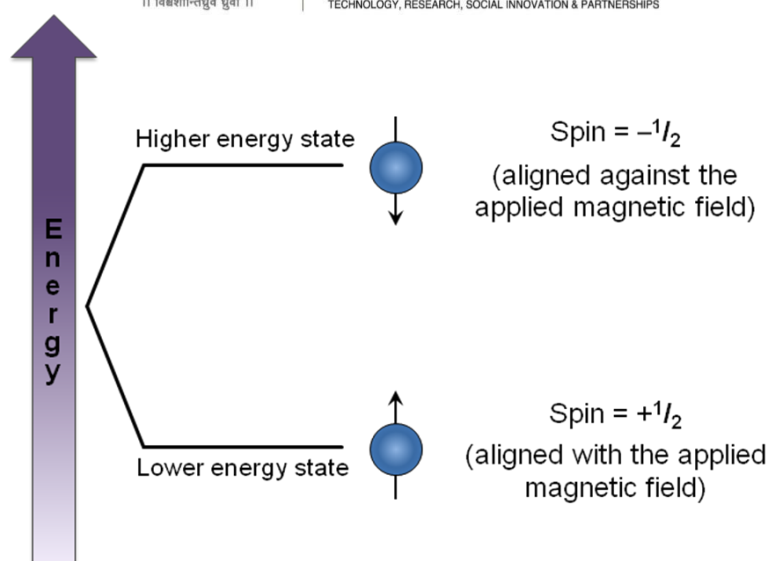


Figure 3 - Effect of external magnetic field on nuclei

An electromagnetic radiation (radio frequency waves) are used to promote the transition between the nuclear energy states. An exact quanta of energy must be used for the transition of nuclei from low energy states to high energy states. The energy difference between the two states corresponds to the energy of the electromagnetic radiation that causes the nuclei to change their energy levels. Whenever the electromagnetic radiation with the specific energy is provided, the nuclei are excited and jump from lower energy states to higher energy states. A nuclear magnetic resonance response to this i.e. relaxation - a free induction decay of the nuclei is observed. The weak signal requires sensitive radio receivers and a Fourier transform of the signal is carried out to extract the frequency-domain spectrum. The location of different NMR signals is dependent on both the external magnetic field strength and the Resonant frequency of electromagnetic radiation. Whenever the strength of the external magnetic field is changed, the resonant frequency of the nuclei will change. In order to eliminate the difference in the NMR signal due to change in strength of magnetic field, an alternative method for characterizing and specifying the location of NMR signals is needed. One method of solving this problem is to report the location of an NMR signal in a spectrum relative to a reference signal from a standard compound added to the sample. Such a reference standard should be chemically unreactive, and easily removed from the sample after the measurement. Also, it should give a single sharp NMR signal that does not interfere with the resonances normally observed for organic compounds. Tetramethyl silane, $(\text{CH}_3)_4\text{Si}$, usually referred to as TMS, meets all these characteristics(17). Since the separation (or dispersion) of NMR signals is magnetic field dependent, one additional step must be taken in order to provide an unambiguous location unit. To correct these frequency differences for their field dependence, we divide them by the spectrometer frequency. The resulting number would be very small, since we are dividing Hz by MHz, so it is multiplied by a million, This operation gives a locator number called the Chemical Shift, having units of parts-per-million (ppm), and designated by the symbol δ . Chemical Shift provides structural information of the compound which is analysed.

$$\text{Chemical shift, } \delta = \frac{\text{frequency of signal} - \text{frequency of reference}}{\text{spectrometer frequency}} \times 10^6$$

The concept of nuclear shielding, which can be detected using NMR through the chemical shifts allows us to determine the structure of a particular compound. Each and every atom is surrounded by electrons which revolve around the nucleus. The revolution of these charged particles creates a magnetic field which influences the nucleus. Thus, any change in the local electronic environment surrounding the nucleus will cause the changes in the magnetic field experienced by the nucleus and thus, will in turn cause changes in the energy levels. This is known as the shielding effect.

The Fourier Transform of the NMR signal helps in spotting out the chemically inequivalent chemical environments caused by shielding of the nucleus. Each peak obtained after Fourier Transform of NMR signal corresponds to the distinct chemical environment. The distinct chemical environment refers to the difference in the number of similar nuclei present in that chemical environment. By calculating the area under the peaks, the number of these similar nuclei in that chemical environment can be determined. Using the number of nuclei the structure of the compound can be deduced (18).

The probe is the part of an NMR spectrometer that does much of the work, in terms of exciting the nuclear spins, and detecting the NMR signal. The probe goes into the centre of the magnetic field, and the sample is inserted into the probe to perform the NMR experiment. The probe contains the radiofrequency (Rf) coils, tuned at specific frequencies for specific nuclei in a given magnetic field. The probe also contains the necessary hardware to control the sample temperature (when combined with an external temperature controller).

Often, probes are constructed with two observe coils, one closest to the sample (the inner coil), and another further from the sample (the outer coil). This allows the probe to respond to multiple frequencies, and to allow the excitation/irradiation of multiple nuclei. The nuclei that use the inner coil are detected with the highest sensitivity. Therefore, a Broadband Observe probe is constructed with the inner-coil tuned to a "broadband" nucleus (i.e. ^{13}C or ^31P), for maximum sensitivity for these nuclei.

The characterization of acidity of zeolites inherently requires the identification of the structure of the zeolites and also the hydroxyl protons which give Brønsted acid characteristics to the zeolites. ^1H magic angle spinning (MAS) NMR has been extensively employed to directly identify hydroxyl protons in microporous zeolites and mesoporous molecular sieves. In this scenario, the bridging hydroxyls (Si-OH-Al) accountable for acidic protons, known as Brønsted acid sites, typically exhibit resonances occurring at proton chemical shifts surpassing approximately 3.6 ppm(16,19). Alternatively, acidity characterization can also be accomplished through the adsorption of appropriate hydrocarbon probe molecules. Regarding sensitivity and resolution of NMR resonance, ^1H

and ^{13}C nuclei, which occur naturally at abundances of 99.98% and 1.07% respectively, and have chemical shift ranges of approximately 20 ppm and 250 ppm respectively, are less desirable compared to the ^{31}P nucleus. The ^{31}P nucleus, with a 100% natural abundance and a chemical shift range exceeding 430 ppm, offers superior characteristics(20). Through the comparison of the conventional methods and ^1H , ^{13}C and ^{31}P NMR for acidity characterization, shown in the Table 1, it can be seen that ^{31}P NMR spectroscopy has the potential to provide the complete information regarding the acid type, site location, number of acid sites and strength of acid sites (21).

Method	Type of Acid Sites		Location of acid site	Number of Acid Sites	Acid Strength
	Brønsted	Lewis			
Titration	✓	✓	✗	✓	✓
TPD (NH_3 Adsorption)	✓	✓	✗	✓	-
IR (Hydroxyls)	✓	✗	✗	✗	✓
IR (Pyridine)	✓	✓	✓	-	-
^1H NMR	✓	✗	✗	-	✗
^{13}C NMR	✗	✗	✗	✗	✓
^{31}P NMR	✓	✓	✓	✓	✓

Legends : (✓) = Can be Identified; (-) = Relation not clear; (✗) = Can't be identified

Table 1- Comparison of the available techniques for acidity

2.6] ^{31}P -NMR for acidity characterization

^{31}P MAS NMR of adsorbed phosphorous molecules, like Trimethyl Phosphine (TMP)(22) and Trialkyl Phosphine Oxides (R_3PO)(23,24) show great sensitivity and reliability in identifying the strength and type of acid sites of solid acid sites. Spectral Deconvolution of the NMR signals along with elemental analysis can provide useful information regarding the distribution and concentration of acid sites(25).

The Brønsted and Lewis acid sites are distributed randomly over the surface of the catalyst due to non-homogeneity during the synthesis of solid acid catalysts. The distribution of the type of acid sites and their respective strength play a very vital role in the catalytic performance of the solid acid catalysts. ^{31}P MAS NMR of adsorbed TMP is more useful in authenticating the acid types (i.e., Brønsted and/or Lewis) and subtle differences in Lewis rather than Brønsted acid strengths in solid acids. As for our reaction of interest, understanding Brønsted acid site distribution and strength is of great importance, alternative method of using Trialkyl Phosphine Oxides (R_3PO) as probe molecules prove to be more practical. Among the Trialkyl Phosphine Oxides (R_3PO) homologues, like Trimethyl

Phosphine Oxide (TMPO), Triethyl Phosphine Oxide (TEPO), Tributyl Phosphine Oxide (TBPO), and Trioctyl Phosphine Oxide (TOPO), a more extensive study on acidity characterization of solid acids has been conducted using TMPO as the probe molecule(26–28).

Trimethyl Phosphine Oxide (TMPO) is an NMR active weak base. As a result, it interacts with the Brønsted acid proton to form TMPOH^+ complex. This TMPOH^+ complex exhibit a wide range of $\delta^{31}\text{P}$ which spans from 50 to 98 ppm depending on the strength of acid sites. The chemical shift of pure TMPO is 39 ppm. Thus, the difference between the chemical shift of TMPO and TMPOH^+ complex is directly proportional to the acid strength of the solid acid catalyst(29). This is primarily because the phosphorus atom is not shielded by the positive charge, as it occurs in phosphines, making it highly sensitive to hydrogen bonding. Depending on the acid strength, the effect of deshielding of electrons varies. This deshielding effect causes downfield (deshielding) chemical shift in the ^{31}P NMR spectrum. Thus, it can be concluded that the strength of the acid catalysts is directly proportional to the chemical shift(30).

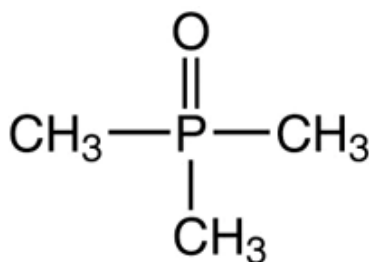


Figure 4 - Trimethyl Phosphine Oxide

2.7] Screening and Design of Catalyst

The purpose of characterization of acidity of solid acid catalyst was to provide an idea about the features of the catalysts which play an important part in the catalytic activity. Through ^{31}P NMR, the type, strength, number and distribution of acid sites was understood. In order to obtain the higher yield of C, it was found out that Brønsted acid sites are required. Instead of screening the catalysts randomly using trial and error method, characterization of acidity of acid catalysts provided a systematic approach to screening of catalysts. After thorough literature review and discussions with the industry regarding the economic constraints, Ion exchange resin (Amberlyst-15) and Zeolites (H-Beta) with varying Si/Al ratio were used for experimentation.

For the future scope of the experimentation and further research, design of novel catalysts either by complete synthesis or modification of the existing solid acid catalysts was investigated in detail. The process for synthesis of a discontinued product (a type of Ion

exchange resin) was studied in detail to produce the Ion Exchange resin which can provide for the reaction. After studying considerable amount of reaction systems like etherification, esterification, epoxidation and Beckmann's rearrangement, importance of mild acidity was understood. The nature and number of weak Brønsted acid sites can be controlled in case of Silicate-1 zeolites. This particular method of design of catalyst relies on modification of Silicate -1 zeolite by introduction of T^{3+} atoms like Al^{3+} , B^{3+} ; creation of mesoporosity in Fe-MFI zeolites and modification using ammonium salts (31). Design and synthesis of Silico Aluminium Phosphates (SAPO) was also studied in detail. The catalyst shows considerable presence of Brønsted acid sites. The morphology, sorption properties and acidity were studied in detail for the catalyst(32).

3) Experimental

3.1] Soaking Experiments

The acylation reaction of interest consists of reactant A, acylating agent B and an solid acid catalyst. The products obtained from the reaction are C and D. In presence of acid catalysts, the reactant A and product C both undergo oligomerization reactions. The oligomerization reaction of pure A and pure C is studied extensively over the years. However, the oligomerization of pure A and pure C with each other is not investigated in detail. This thought led us to design the soaking experiments. As the oligomerization reactions occur in the reaction of our interest it is critical to know the degree of oligomerization of each and every compound. It is also crucial to know which compound is responsible for formation of large chain oligomers which primarily cause the deactivation of the catalyst. It is very well known fact that compound A undergoes self-oligomerization reaction. It is also common knowledge that highly reactive compound C also oligomerizes with itself. However, we also suspected that compounds A and C also cross oligomerize giving rise to large chain oligomers. Thus, the experimental design of the soaking experiments was done in such a way that we could test the individual and cross oligomerization of A and C in presence of compounds B and D.

There are a total of 4 chemical species in the reactor system – 2 reactants (A and B) and 2 products (C and D). In order to study, the interactions of these 4 species with each other, we prepared binary mixtures of the reactants and products and added the acidic catalyst to see the interactions. There 4C_2 combinations possible for obtaining binary mixtures of 4 chemical species. The 6 (4C_2) Soaking Experiments were thus planned as follows :

Sr. No,	Reaction ID	Components
1	SE-1	A + D
2	SE-2	C + D
3	SE-3	B + C
4	SE-4	A + C
5	SE-5	A + B
6	SE-6	B + D

Table 2 - Soaking Experiments

The SE -6 experiment which consisted of reactant B and co-product D was not performed as the reaction between B and D is well reported and does not cause any deactivation of catalyst.

The binary mixtures were first added to a 50 ml 3 necked Round Bottom flask. A two -pass condenser was used in order to avoid the vapour loss of the volatile component A. A chiller with a mixture of ethylene glycol and water was used to maintain the temperature of the condenser at -3°C . The glass reactor was kept in an oil bath and heated with constant stirring of 300 rpm. As the temperature of the reaction mixture reached the desired set point

of 60 ° C, the catalyst was added to the reactor. The catalyst used in the reaction was Amberlyst-15 Ion Exchange Resin. We chose ion exchange resin as the catalyst over zeolite specifically to observe the effect of strength of acid sites on the reactions between the chemical species. Amberlyst-15 is a strongly acidic ion exchange resin with strongly acidic Bronsted acid sites. Zeolites on the other hand have both strong and weak Bronsted and Lewis acid sites depending on the Si/Al ratio.

After completion of 60 mins, the reaction mixture was cooled down to room temperature. The product mixture obtained after the individual soaking experiments was preserved under cold conditions to ensure zero loss of the product or unreacted reactants. The catalyst samples were then washed thoroughly using 30 ml Dichloromethane (DCM) to remove the adsorbed reactants and products. The catalyst was then dried at 100 °C for 24 hours and stored. The product mixtures obtained from the soaking experiments were sent for High Resolution Liquid Chromatography Mass Spectroscopy (HRLCMS) to identify the oligomerization products. Liquid chromatography Mass Spectroscopy (LCMS) is a technique which combines high performance liquid chromatography HPLC, a powerful analytical separation technique with mass spectroscopy ,a powerful analysis & detection technique. There are two common atmospheric pressure ionization (API) LC/MS process: Electrospray Ionization (ESI) & Atmospheric Pressure Chemical Ionization (APCI). Both are soft ionization technique. Agilent iFunnel technology combines highly efficient electrospray ion generation and focusing of Agilent Jet Stream technology with a hexabore capillary sampling array and dual-stage ion funnel for increased ion sampling and transmission. NanoHPLC combined with mass spectrometer can analysed small molecule as well large molecules like proteins High Resolution Liquid Chromatograph Mass Spectrometer developed by Agilent Technologies, USA was used for analysis.

The catalyst samples were sent for ^{31}P – NMR Spectroscopy to analyse the number of active sites and strength of acid sites. A control case was maintained for the catalyst samples where fresh catalyst was taken to get a comparative view of the catalysts in terms of the deactivation.

3.2] Deactivation of catalyst Experiments

Through the Soaking Experiments we were able to identify the interactions of reactants and products with each other in presence of the catalyst. The study of the spent catalyst of soaking experiments though gave us an exhaustive view of the deactivation caused by the reaction between the reactants and the products on a strong acid catalyst, the experiments did not provide us with the insights related to the deactivation of catalyst with time.

Therefore, we planned the deactivation of catalyst experiments to study the deactivation of catalyst with time. Unlike Soaking experiments, we took only the feed mixture of reactants

(A and B) to study the deactivation of catalyst. In these set of experiments too, Amberlyst-15 was used as the catalyst. The use of strongly acidic ion exchange resins helped us greatly to observe the oligomerization of reactants and products in presence of strong acid catalyst. The results obtained from these were later on compared with the batch reactions carried out using zeolites. Unlike ion exchange resins, zeolites have both strong and weak Lewis and Bronsted acid sites. The proportion and presence of these acid sites depends on the Si/Al ratio.

To study the deactivation of the catalyst with time, the reaction mixture consisting of reactants A and B in molar ratio 1:1.2 was added to a 50 ml 3-necked Round bottom flask. It was found out through the soaking experiments, that the product C also reacts actively with reactant B. Thus, in order to compensate for the side reaction consumption of the reactant B, reactant B was taken in 20% excess. The reactant A being a volatile compound, a two-pass condenser was used to avoid the loss of reactant A during the reaction. Using a mixture of ethylene glycol and water the temperature of the condenser was maintained at -3° C. The reactor was heated to maintain the temperature of the reaction mixture at 60 °C along with constant stirring of 300 rpm. 2g of catalyst (Amberlyst-15) was then added to the reaction mixture. As the main purpose of the set of experiments is to identify the deactivation of catalyst with time, the reaction was arrested at specific times. A total of 5 deactivation of catalyst experiments were performed where the reaction was arrested at different times (Table. 1). As soon as the reaction was stopped at the specific time, the reactants were drained off immediately and the catalyst samples were washed with 30ml DCM to remove the adsorbed reactants from the catalyst surface. The catalysts obtained from the deactivation of catalyst experiments were sent for ³¹P- NMR spectroscopy analysis to obtain insights on decrease in number of active sites with respect to time and the strength of these acidic sites.

Sr. No.	Reaction ID	Time (mins)
1	DC 1	1
2	DC 2	5
3	DC 3	10
4	DC 4	25
5	DC 5	45

Table 3 - Deactivation of catalyst experiments

3.3] Preparation of ³¹P-NMR samples

A standard operating procedure was maintained for preparation of ³¹P-NMR samples. The catalyst samples of the Soaking experiments and Deactivation of catalyst experiments which were washed with DCM were calcined at temperature of 300° C to remove the moisture and

physiosorbed reactant and product species. The number of acid sites present in 2g catalyst sample were calculated. The amount of TMPO to be added so that each and every acid site of the catalyst could be identified was calculated. A 20% excess of TMPO was taken to account for the losses. TMPO is a highly hygroscopic molecule and thus cannot be exposed to the atmosphere. Dry DCM was obtained by distillation and complete removal of any residual moisture. 20 g of TMPO was then added to 50 ml of Dry DCM in a glove box and the mixture was then stored at ambient conditions for future use. Precisely, 0.2 g of this TMPO + Dry DCM was added to the 0.2 g samples of Soaking experiments and deactivation of catalyst experiments. Duplicate samples of each and every case of Soaking experiments and deactivation of catalyst experiments were prepared and same amount of TMPO + Dry DCM was added to the samples. The duplicate samples ensured the repeatability of the results and also allowed to measure the systematic or the manual error through statistical analysis. The samples were left for evacuation inside the hood at 50° C to remove the DCM and to disperse the added TMPO. The samples were then heated at 165° C for 1 hour. The dried samples were then sent for ^{31}P NMR facility.

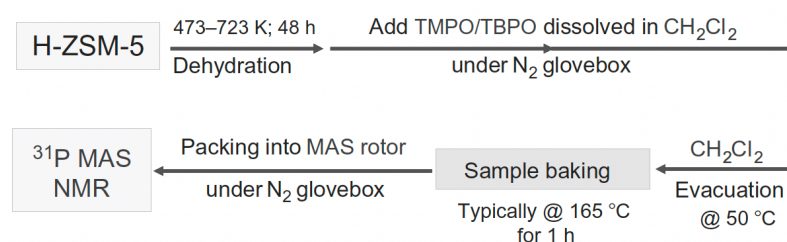


Figure 5 - ^{31}P NMR sample preparation(33,34)

3.4] Batch reactions

We conducted the batch reactions for the acylation reaction of industrial importance. The batch reactions were carried out to investigate the effect of operating conditions on the reaction yield. Batch reactions were important for getting valuable insights into the reaction system and helped for developing an industrial scale process for production of C.

A 100 ml Stainless Steel Autoclave was used for carrying out the batch reactions. For the batch reactions, 100g of reaction mixture was charged as feed. The molar ratio of feed was A : B = 1 : 1.2. Reactant B was taken in 20 % excess in order to account for the consumption of B in the side oligomerization reactions with product C. The acylation reaction of A with acylating agent B is well reported at temperatures ranging from 40 – 60 ° C. We carried out the batch reactions in the SS Autoclave at higher temperatures. Our hypothesis for carrying out the reaction at higher temperatures was that, at higher temperatures, the rate of reaction increases and thus the productivity can be increased greatly. The other experiments carried out on the autoclave included the effect of catalyst loading on the conversion and yield and

the effect of addition of co-product D into the feed to avoid the oligomers and catalyst from sticking to the walls of the reactor and reduce the effort and time required for the cleaning procedures.

The products of the batch reactions performed in Autoclave were sampled at specific times using a sampler developed by us for the specific operation. The reactants and products were analysed using Gas Chromatography. Gas chromatography (GC) is an analytical technique used to separate and detect the chemical components of a sample mixture to determine their presence or absence and/or quantities. GC uses a carrier gas in the separation, this plays the part of the mobile phase. The carrier gas transports the sample molecules through the GC system, ideally without reacting with the sample or damaging the instrument components.

Gas chromatograph developed by Agilent Technologies, USA equipped with a liquid autosampler was used for analysis of the reactants and products Internal calibration method was used to correct the variations in sample injection and chromatographic separation. By adding a known amount of a reference compound (called an internal standard) to your sample, you can compare its signal with the signal of your analyte of interest and calculate the concentration of the analyte based on a calibration curve. The internal standard should have similar chemical and physical properties to the analyte, but should not interfere with its detection or quantification. The benefits of using the internal calibration method are well documented and impactful for obtaining high quality and reliable results(35).

3.5] Fixed Bed Reactor

It was found that if the life of the catalyst was increased such that the catalyst would allow 90 % conversion of the reactants for a long period of time, the process could be easily scaled up and could be more profitable as opposed to the existing liquid acid process. The products and reactants also undergo oligomerization reactions in presence of acid catalysts. In order to tackle this issue, we designed a continuous reactor setup. The acid catalyst extrudes were packed into a column. The reactants were sent into this catalyst bed and the products and unreacted reactants were withdrawn from the other end of the column.

The jacketed reactor length was 35 cm with volume of 65 ml. The jacketed reactor was mounted on a iron frame. Hot oil was circulated through the jacketed reactor to maintain the temperature of the reactor. A feed tank of 5kg was setup on a weighing balance to store the feed. The weight of the feed tank was continuously monitored using the weighing balance for mass balance calculations. A peristaltic pump was used to pump the feed into the fixed bed reactor. The feed flow rate was varied from 0.2 ml/min to 10 ml/min. A back pressure regulator was fixed at the end of the reactor to maintain the pressure at the outlet at 6 bar. A condenser was connected in order to cool down the product stream and unreacted reactants. A sampler was also attached to the condenser which allowed us to withdraw samples at specific time intervals to monitor the conversion of reactants and yield of target product C.

The collected products were analysed using Gas Chromatography for tracking the conversion and yield.

The design of experiments was done using Box Behnken Design (BBD) and Response Surface Methodology (RSM). For the designing the experiments using this method, variables X_1 , X_2 and X_3 were defined for temperature, Feed molar ratio and feed flow rate.
 X_1 = Temperature ($^{\circ}\text{C}$)

X_2 = Feed Molar Ratio (B : A)

X_3 = Feed Flow rate (ml/min)

Each of these variables X_1 , X_2 and X_3 are placed at one of the three equally spaced values which are usually coded as -1, 0 and 1. The design was fitted to quadratic model with $R^2 = 0.99$ thereby establishing statistical significance of the experimental design.

The lowest value for temperature was taken as 45°C and the highest temperature was taken as 75°C . Thus, the coded -1 corresponded to 45°C , 0 corresponded to 60°C and 1 corresponded to 75°C .

Similarly values were fixed for feed molar ratio and feed flow rate. For molar feed ratio, the three equally spaced values chosen were 1, 3.75 and 6.5. Feed flow rate values were varied from 4 ml/min to 10 ml/min with 7 ml/min being the intermediate value.

Thus, X_1 , X_2 and X_3 could be represented as :

$$X_1 = \{ 45, 60, 75 \} = \{ -1, 0, 1 \}$$

$$X_2 = \{ 1, 3.75, 6.5 \} = \{ -1, 0, 1 \}$$

$$X_3 = \{ 4, 7, 10 \} = \{ -1, 0, 1 \}$$

The Box Behnken Table was then formulated using the three coded and uncoded values.

Sr. No	Coded			Uncoded		
	X1	X2	X3	X1	X2	X3
0	-1	-1	0	45	1	7
1	1	-1	0	75	1	7
2	-1	1	0	45	6.5	7
3	1	1	0	75	6.5	7
4	-1	0	-1	45	3.75	4
5	-1	0	-1	45	3.75	4
6	1	0	-1	75	3.75	4
7	0	0	1	60	3.75	10
8	0	0	1	60	3.75	10
9	0	-1	-1	60	1	4
10	0	1	-1	60	6.5	4
11	0	-1	1	60	1	10
12	0	1	1	60	6.5	10
13	0	0	0	60	3.75	4
14	0	0	0	60	3.75	4
15	0	0	0	60	3.75	4

Table 4 - Box Behnken Experimental Design

It can be seen through the Box Behnken Experimental Design table that the runs 7 and 8 were the repeatability runs included to increase the confidence on the experimental results and also test the repeatability of the experiment. The FBR BBD 7 run was conducted on the Fixed bed reactor at a small scale. We carried out the FBR BBD 8 run at pilot scale wherein the reactor was operated for 22 hours and the conversion of A and yield of C was noted. The experiments 13,14 and 15 were also the repeatability runs.

A total of 14 experiments were planned for the fixed bed reactor system using the Box-Behnken designs (BBD). 3 experiments of the Fixed Bed Reactor Box Behnken Design (FBR-BBD) were performed by me in the course of my internship.

3.6] Distillation

The product mixture consisted of product C, co-product D, unreacted reactants A and B and the oligomers of the reactant A and product C. For obtaining pure product C, the separation process used was distillation. The boiling points of the compounds have a significant difference, making distillation the most preferred method for separation of the product from the mixture.

Sr. No.	Component	Boiling Point (°C)
1	A	31
2	B	139.5
3	C	168
4	D	118
5	Oligomers	>200

Table 5 - Boiling points of components

The batch distillation was setup using a 250ml 2-necked Round Bottom flask, condenser and a round bottom flask for product collection. The 250 ml Round bottom flask was kept in a oil bath. The temperature of the mixture to be distilled was monitored continuously using a thermocouple. The temperature of the vapour was also noted to understand the vapour load and composition of the vapour entering the condenser.

The target product has a boiling point of 168 °C. Therefore, the temperature of the mixture was maintained at 165 °C. The low boiling point compounds namely A, B and D were recovered in first D1 cut of the distillation. The temperature was then increased to 180 °C. The target product C was obtained in the collection flask. The sticky oligomers with high boiling points remained in the reboiler flask and needed to be cleaned with strong acid. The purity of the products obtained was also not high enough to continue the operation.

Therefore, in order to increase the separation efficiency, a fractionating column was added to the distillation setup. The diameter (D) of the fractionating column was 2 cm. For efficient separation, (Column Diameter/ Packing Diameter) > 8 to 12. Thus, Random ceramic packings with diameter D₁ were used in the fractionating column. The diameter of the fractionating column was 4 cm. Thus, the random packings were selected with diameter D₁ where ,

$$D_1 = D / 10 = 0.4 \text{ cm.}$$

The random packings increased the surface area over which the components can interact and facilitated efficient heat transfer thereby greatly increasing the separation efficiency. The use of a fractionating column however, required us to increase the heat duty of the reboiler. This was primarily because a temperature drop of ~ 10 °C was observed throughout the fractionating column. The use of fractionating column thus greatly increased the time and energy required for the distillation operation.

In order to perform the distillation effectively in a short duration of time, we used vacuum. The reduced pressure brought down the boiling points of the components by a certain degree thereby helping us to limit the required heat duty and also save time.

The change in boiling points of the components due to use of vacuum needed to be known in order to separate the components. Therefore, we plotted the Vapour-Liquid Equilibrium curves at different pressures. The components selected for plotting the VLE curves at

different pressures were B and C. B was chosen specifically because it is the highest boiling point component among the unwanted low boiling point products of the distillation. The C being the target product, we needed to know the exact boiling point of the compound at different pressure so that we could adjust the heat duty to obtain pure product C. The oligomers being very high boiling point compounds, were not included in the VLE plots. The VLE plots provided a great insight into the cut-off temperature at which the components A, B and D could be obtained. For plotting the VLE curves, the pressure was decreased from 1.0132 bar to as low as 0.0033 bar. The pressure of 1.0132 bar corresponded to atmospheric pressure or the case of no vacuum. The pressure of 0.0033 bar corresponds to the case where full vacuum can be provided.

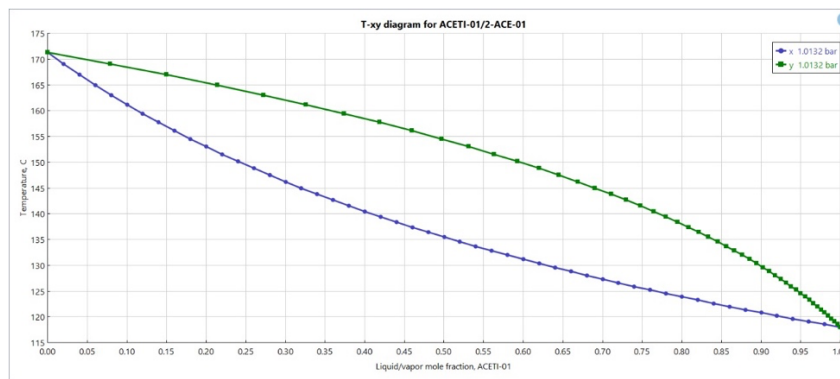


Figure 6 - VLE Curve of B and C at 1.0132 bar

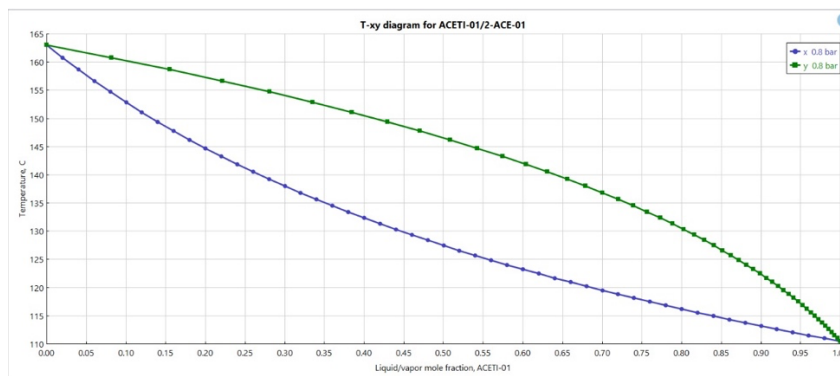


Figure 7 - VLE Curve of B and C at 0.8 bar

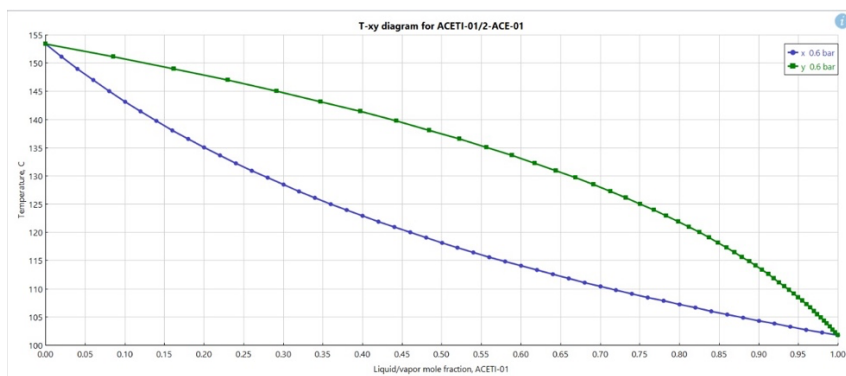


Figure 8 - VLE Curve of B and C at 0.6 bar

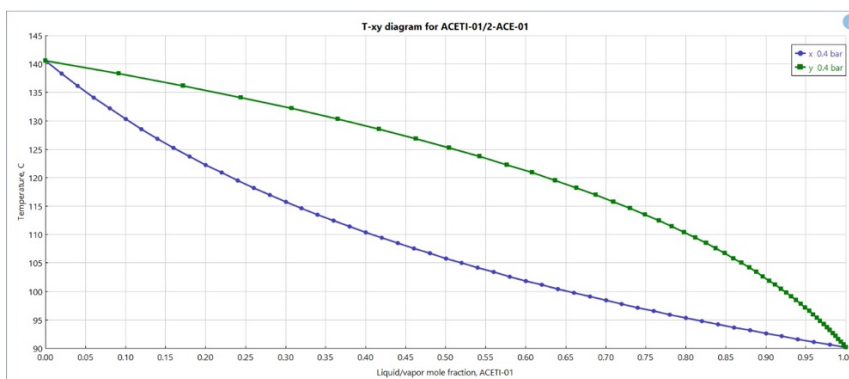


Figure 9 - VLE Curve of B and C at 0.4 bar

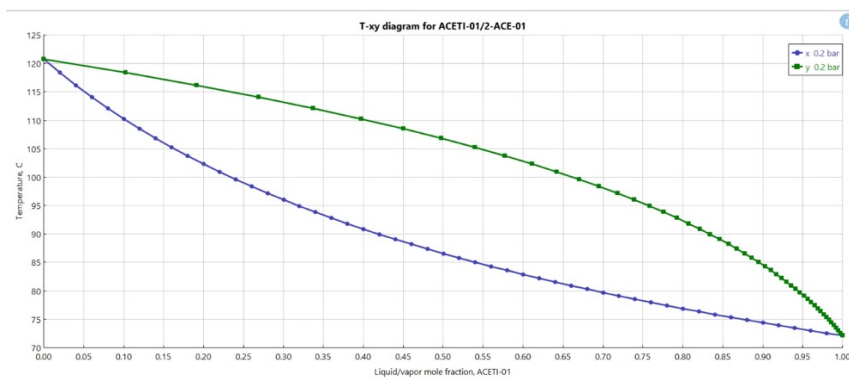


Figure 10 - VLE Curve of B and C at 0.2 bar

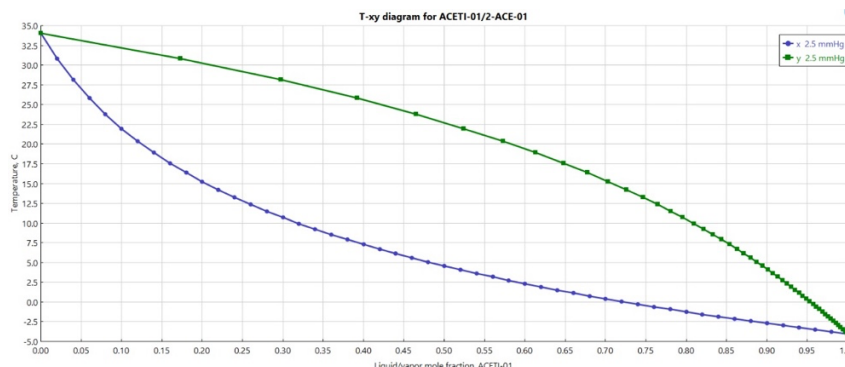


Figure 11 - VLE Curve of B and C at 0.0033 bar

Through these VLE plots, it was found that at very low pressures, the separation efficiency of the components can decrease greatly as the boiling points are very close to the room temperatures. By studying the VLE plots carefully and the capacity of the vacuum pump available to us, we decided to operate the distillation operation at the reduced pressure of 0.2 bar. The boiling point of B at this pressure was found to be $\sim 72^\circ\text{C}$. The cut-off was thus, fixed at 120°C . The cut-off point was set after taking into consideration the temperature drop across the column and the elevation in boiling points of the residual mixture as the distillation mixture became leaner in low boiling point compounds. The first cut (D1), thus, consisted of components A, B and D. As the cut-off point was reached, the temperature was ramped up to the next set point of 140°C which was well above the boiling point of component C at that pressure. The component C was collected as the second cut (D2). A small fraction of C and the oligomers were left in the reboiler vessel which were sent for HRLCMS to study the effect of high temperature and low pressure on the structure of oligomers.



Figure 12 - Distillation Setup

4) Results and Discussion

4.1] Soaking experiments

The catalyst samples of the soaking experiments and the control case were analysed using ^{31}P -NMR. The ^{31}P -NMR gave us useful insights regarding the number of active sites after deactivation and the strength of acid sites.

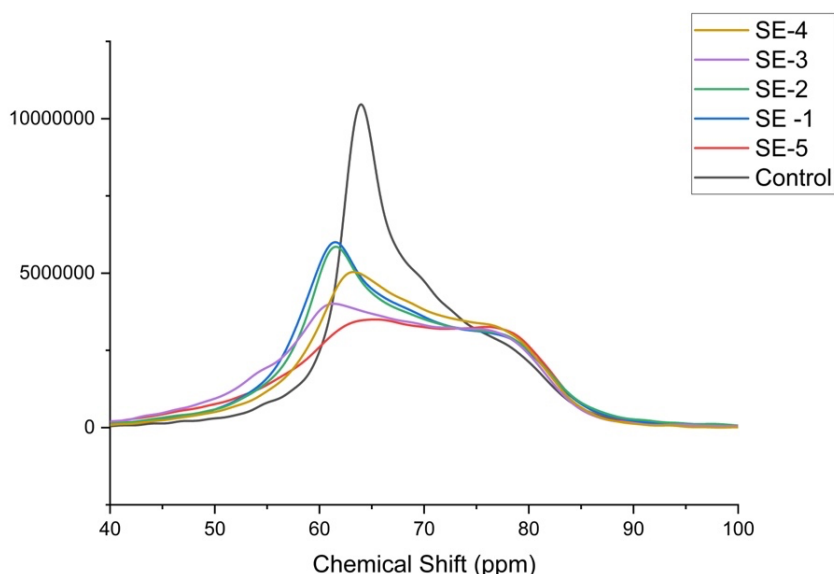


Figure 13 - ^{31}P -NMR of SE + Control

The above figure shows the Area Vs Chemical Shift trend of the Soaking Experiments along with the control sample. The height of the peak corresponding to 63-64 ppm can be seen decreasing in SE1, SE2, SE3, SE4 and SE5. The peak height is seen to maximum in case of control which is basically the fresh catalyst. The height of the peak has decreased significantly in SE 5 case. The decrease in peak height generally signifies the decrease in number of acid sites. The decrease in number of active sites is caused by deactivation of catalyst due to deposition of oligomers.

The ^{31}P -NMR spectra of the Control, SE1, SE2, SE3, SE4 and SE5 were individually analysed. During the analysis the peaks were deconvoluted to get a clear idea about the individual peaks corresponding to specific chemical shifts.

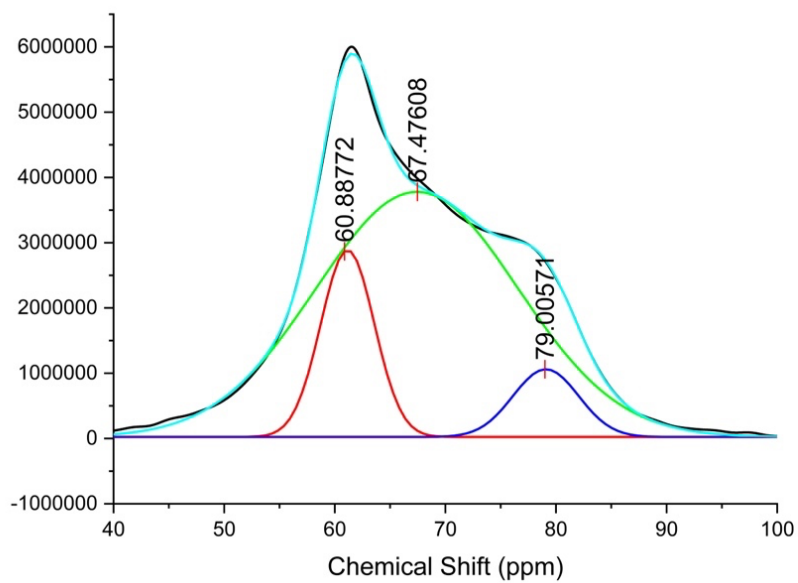


Figure 14 - ^{31}P -NMR of SE-1

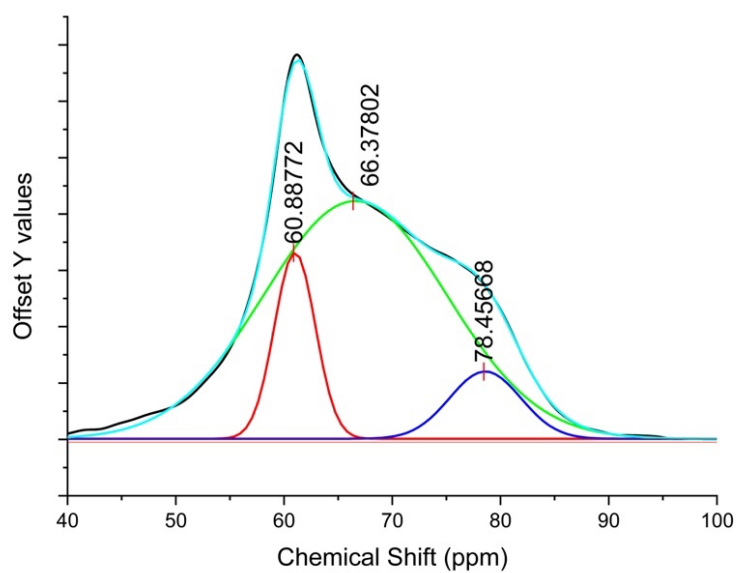


Figure 15 - ^{31}P -NMR of SE-2

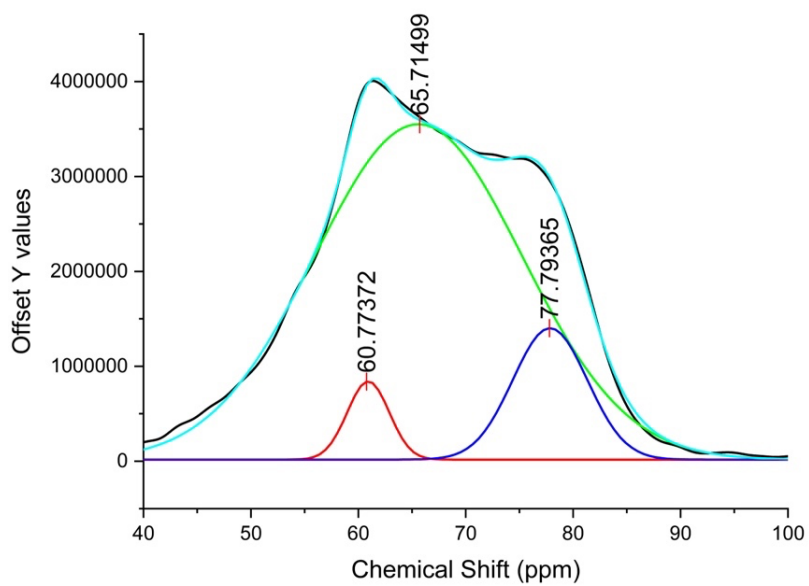


Figure 16 - ^{31}P -NMR of SE-3

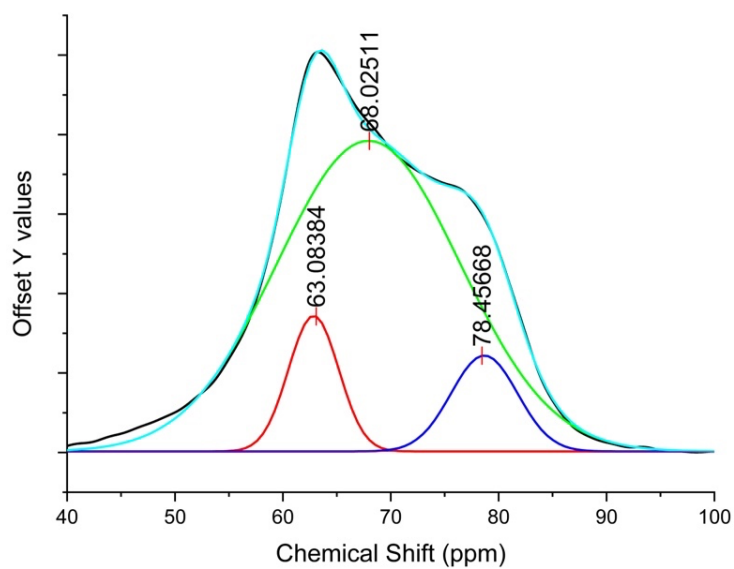


Figure 17 - ^{31}P -NMR of SE-4

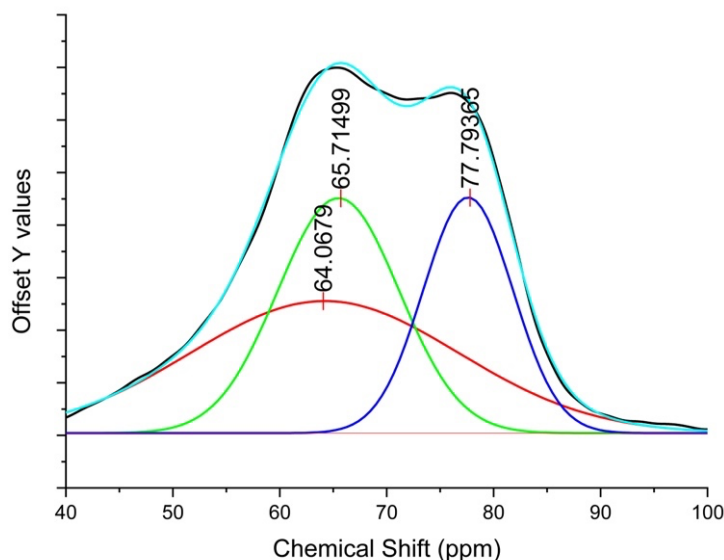


Figure 18 - 31 P-NMR of SE-5

The figures show the deconvoluted peaks. Three major peaks were obtained for Control, SE1, SE2, SE3, SE4 and SE5 samples. The peaks obtained by Gaussian deconvolution were categorized as relatively weak acidic, medium acidic and highly acidic. The weak acidic sites corresponded to peaks with chemical shift in the range 60 – 64 ppm. Peaks with chemical shifts in the range 65 - 69 ppm were categorized as medium acidic sites. The peaks with chemical shift 77 - 78 ppm were the highly acidic sites. For better understanding the trend of the change in number of active sites of specific strength of acids, the area under each peak was calculated. This area corresponded to the number of active sites available with the catalyst after the completion of the reaction. For ease of interpretation of the data, the areas of the medium and highly acidic sites corresponding to the peaks 65-69 ppm and 77-78 ppm were added. This was done primarily to split the acid sites into two broad categories – weakly acidic and highly acidic. We plotted comparative plots of Control, SE1 SE2, SE3, SE4 and SE5 for (Medium + High) acidic sites and weakly acidic sites

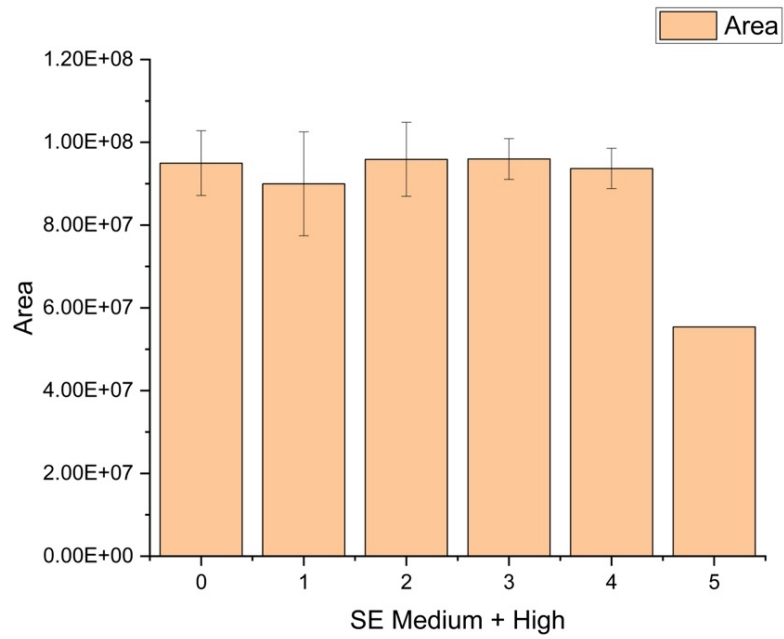


Figure 19 - Comparison of Number of active sites of medium and high acidity

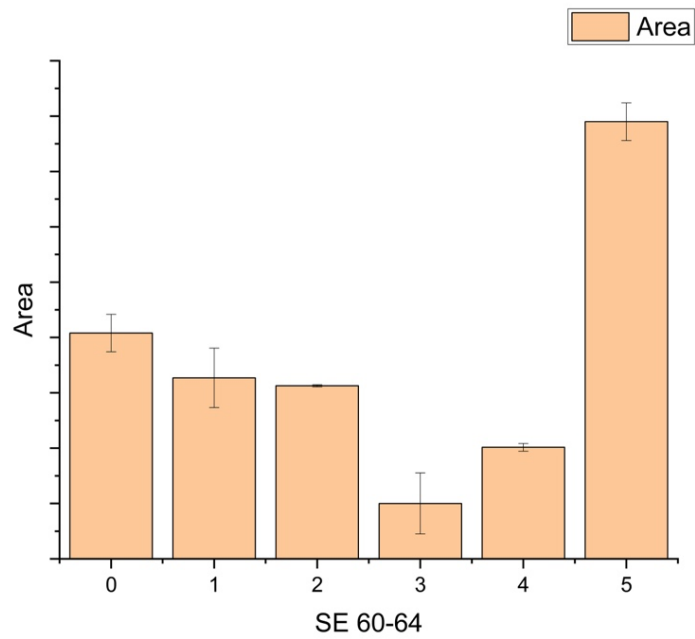


Figure 20 - Comparison of Number of active sites of low acidity

Through the comparative graphs it can be seen clearly that the number of medium and highly acidic sites do not undergo a significant change. The number of acid sites can be seen to remain constant. The error bars confirm that the change in number of acid sites for medium and highly acidic sites for SE1, SE2, SE3 and SE4 is statistically insignificant. However, a very different trend is observed in the case of SE-5. The number of medium and highly acidic sites is seen to decrease significantly. We can therefore conclude that the in presence of reactants A and B, the oligomerization reactions are catalysed by the medium and highly acidic sites. The oligomerization reactions take place at medium and highly acidic sites and the large chain oligomers which are formed as products reduce the number of effective active sites deactivating the catalyst at a great extent.

For sites with low acidity, corresponding to chemical shift of 60 – 64 ppm, the area under the peaks and thus the number of active sites decreases significantly for SE-2, SE-3 and SE-4 samples while an significant increase in number of low acidic sites is seen for the SE-5 sample. A change in area can also be observed in SE-1, however, after careful investigation and study of the error bars it can be understood that the change is statistically insignificant. For SE-1, there is no change in area and thus number of low, medium or highly acidic sites. It can be therefore said that there is no reaction between reactant A and product D which produces oligomers responsible for deactivation of catalyst. For SE-2, SE-3 and SE-4 there is a net decrease in the number of sites with low acidity. We can thus conclude that the oligomerization reactions between C and D, B and C & A and C are catalysed by sites with low acidity. These oligomerization reactants produce long chain products which block the active sites with low acidity. In the case of SE-5, the oligomerization reactions are catalysed by the medium and highly acidic sites. The oligomers which get attached to the medium and highly acidic sites convert these sites into weakly acidic sites. Therefore there is apparent increase in the number of the weakly acidic sites in SE-5 and a decrease in medium and highly acidic sites.

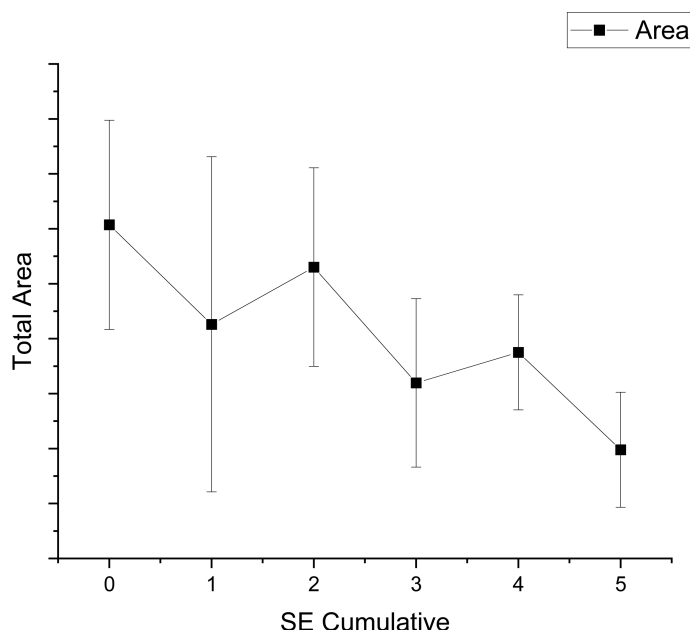


Figure 21 - Comparison of total number of active sites

The mother liquor obtained after the separation of catalyst was analysed using HRLCMS. This was done primarily to identify the oligomers formed during the course of the reaction. The identification of oligomers was done analysing the structure of each and every possible oligomer. The structure of the possible oligomers was then analysed and the exact mass of the compound was found out. The exact mass of the possible structure was then matched with the mass obtained from the results of HRLCMS. If the value of the exact mass matched with that obtained from the HRLCMS data, the structure of the oligomer was accepted to be final. The error margin for the matching of the exact mass to the molecular weight of the compound obtained through HRLCMS was measured in ppm (parts per million). Parts per million is a unit of measurement used to express the mass resolution or the mass accuracy of the mass spectrometer. The ppm value measures the approximation error in the measured mass compared to the expected or the theoretical mass.

Formula used for calculation of ppm is :

$$ppm = \frac{(Theoretical\ M.Wt.\ value) - (Experimental\ M.Wt.\ value)}{(Theoretical\ M.Wt.\ value)} \times 10^6$$

The accepted value of the measured mass should be within 5 ppm of the exact mass of the compound for the structure to be a perfect match.

Sr. No.	HRLCMS M.Wt.	Exact Mass	Ppm error
1	110.0367	110.0368	0.90
2	152.047	152.0473	1.97
3	270.088	270.089	3.70
4	202.062	202.063	2.96
5	312.098	312.099	3.84

Table 6 - Identified Oligomers and ppm

A total of 5 oligomers were identified whose molecular weight matched exactly with the results of HR-LCMS. 2 large chain oligomers (M.Wt. >500 units) remained unidentified. One of these unidentified oligomers was obtained from the product mixture of SE-1 reaction. Thus, it can be concluded that the large chain oligomer was a long chain compound and a product of self-oligomerization of reactant A. The oligomer corresponding to Molecular Weight 202.0630 was the predominant compound found in the product mixtures of SE-2, SE-3, SE-4 and SE-5. This shows that the oligomerization of product is more than that of the reactant.

4.2] Deactivation of Catalyst

In order to observe the deactivation of catalyst at different time slots, the catalysts obtained from the DC experiments were analysed using ³¹P- NMR. ³¹P-NMR was also done for fresh catalyst sample which was maintained as the control case. The use of control case helped us compare the deactivated catalyst with the fresh catalyst and determine the number of active sites present in the catalyst sample after deactivation.

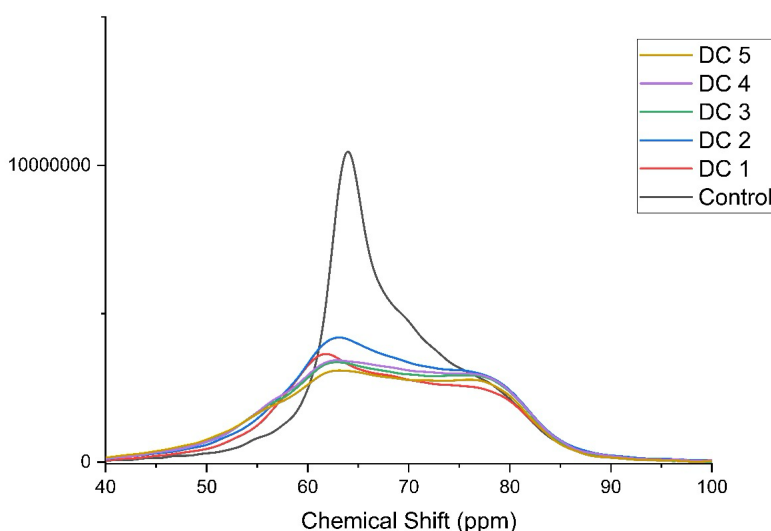


Figure 22 - ³¹P-NMR spectra of Deactivation of catalysts combined

The figure shows the ^{31}P -NMR results of Control and Deactivation of catalysts experiments – DC1, DC 2, DC 3, DC 4 and DC 5. Through the figure, it can be clearly seen that the peak height corresponding to chemical shift of 63-64 ppm is decreasing.

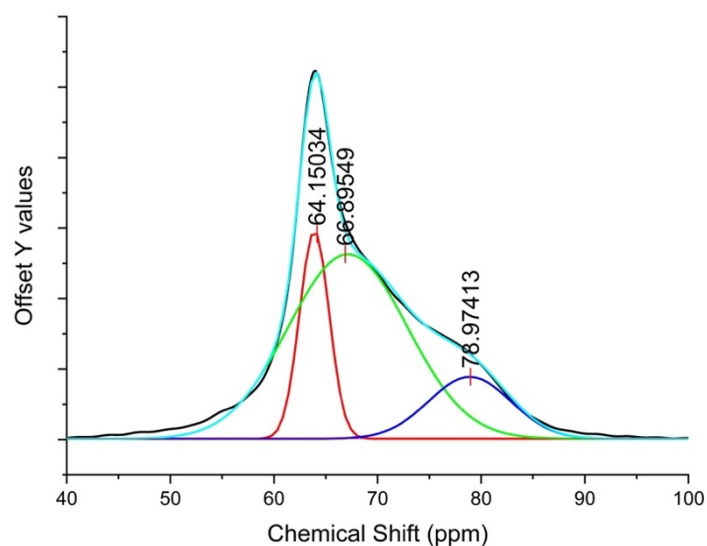


Figure 23- ^{31}P -NMR spectra of Control

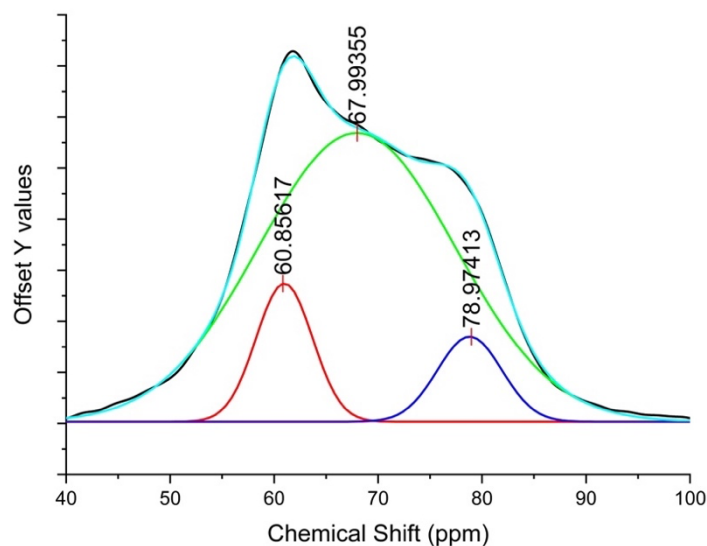


Figure 24- ^{31}P -NMR spectra of DC 1

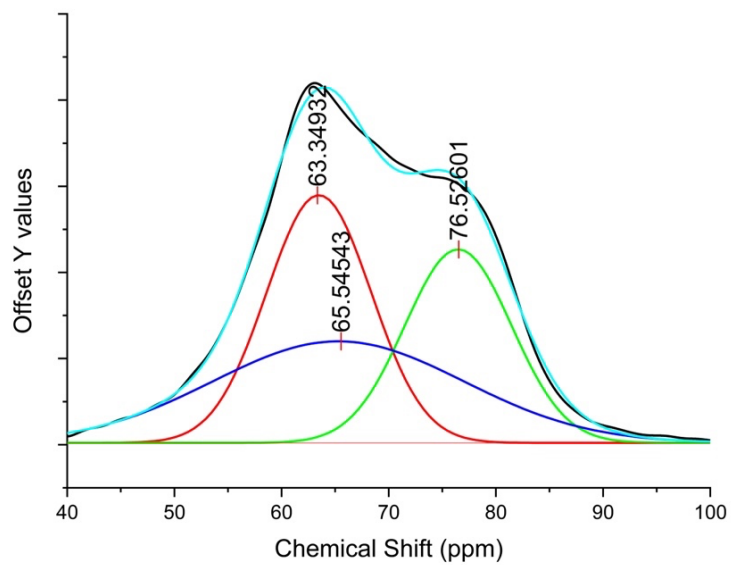


Figure 25 - ³¹P-NMR spectra of DC 2

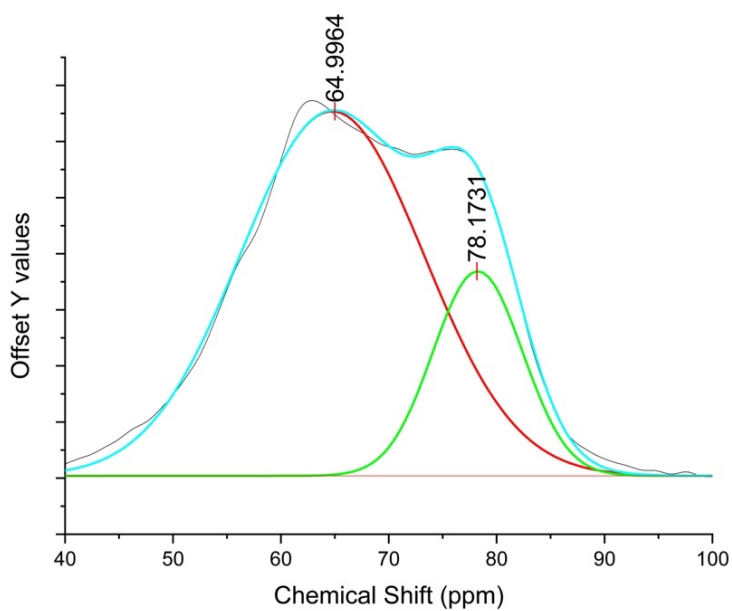


Figure 26- ³¹P-NMR spectra of DC 3

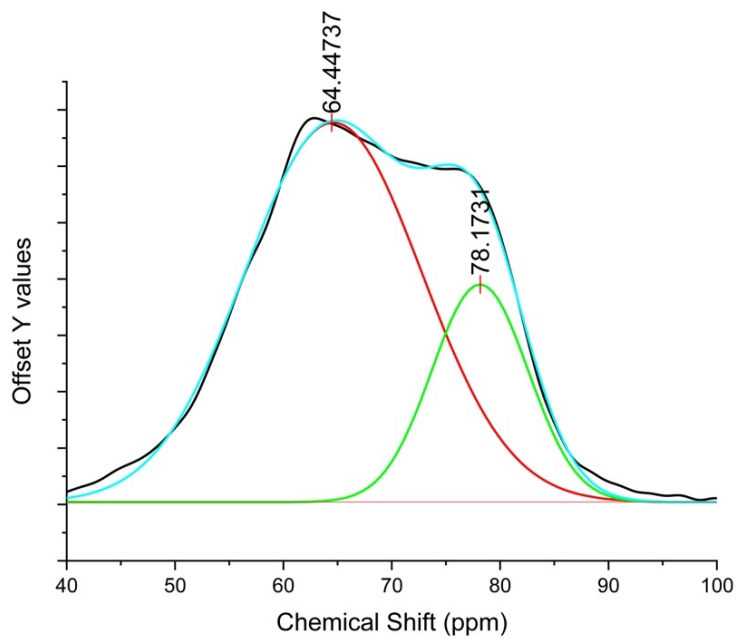


Figure 27 - ³¹P-NMR spectra of DC 4

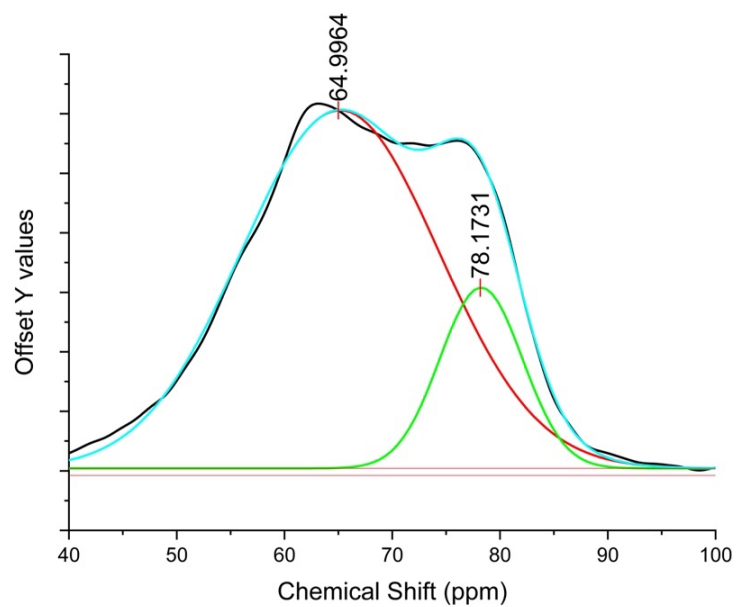


Figure 28- ³¹P-NMR spectra of DC 5

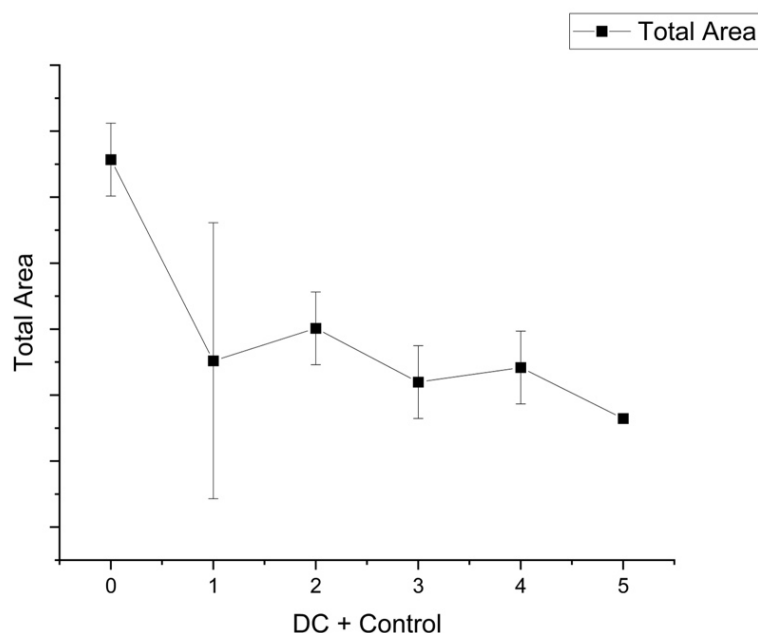


Figure 29 - Total Area of DC Experiments

The control case is maintained as a reference case which allows for comparison of catalysts for deactivation. The total area under the deconvoluted peaks of ^{31}P NMR spectrum of DC 1 - 5 was calculated. The trend shown in the Figure 29 makes it clear that with respect to time, the total area decreases. The total area corresponds to the number of active sites. The number of active sites is the highest for the control case which is the fresh catalyst. The decrease in the number of active sites signifies the severe deactivation of catalyst. The data proves to be of great importance as it can be concluded from the trend that the deactivation occurs almost instantaneously at $t = 1$ min. The Ion exchange resin (Amberlyst-15) contains majority of strong acidic sites. This data can also help to infer that the rate of oligomerization reaction and thus deactivation of catalyst is higher in case of highly acidic Brønsted acid sites.

4.3] Batch Reactions

The acylation reactions of reactant A in batch mode were studied previously using different Ion Exchange Resins. In the present study, we studied the acylation reaction using Zeolite as the catalyst. The zeolite used was H-Beta 300 zeolite with Si/Al ratio of 300.

4.3.1. Effect of temperature

The acylation reactions were carried out using H-Beta 300 Zeolite at high temperatures ($> 120^\circ\text{C}$).

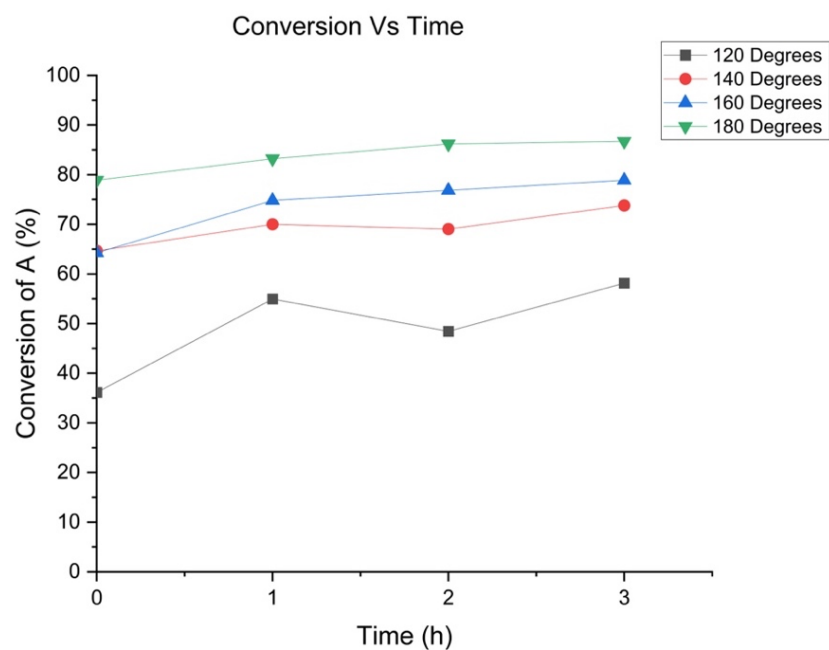


Figure 30- Conversion of A with Time at High T

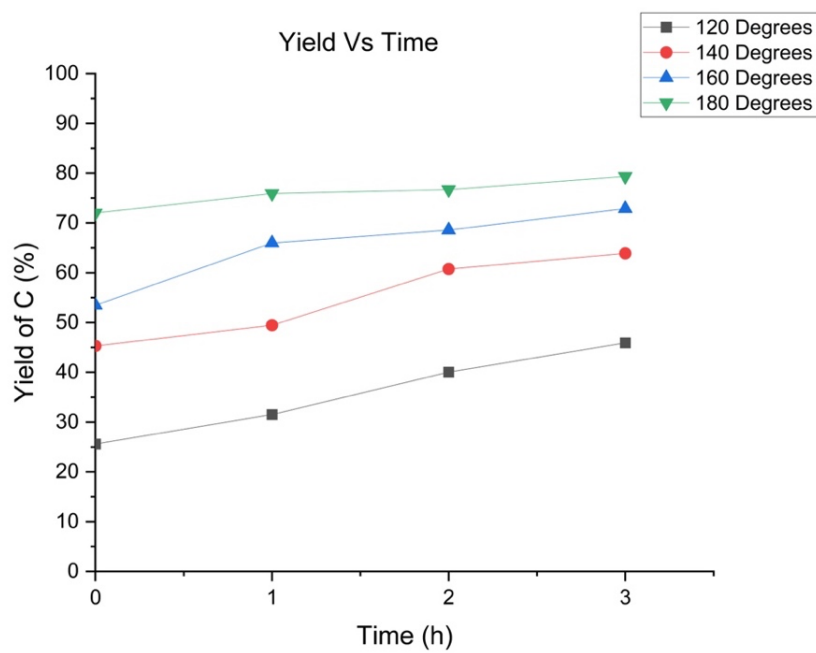


Figure 31 - Yield of C with Time at High T

The conversion of reactant A and yield of product C are both seen to increase with an increase in temperature. Thus, temperature of 180 ° C was selected as the operating Temperature for the process of production of C. However, increase in temperature also caused some challenges which needed to be tackled for better process control and operation. As the temperature was increased, the suspended catalyst and the oligomers of the product and reactant were found to stick to the reactor walls. This caused several problems as it increased the time and effort required for cleaning the reactor and also increased the loss of catalyst.

In order to tackle the problems associated with the increase in Temperature we investigated the effect of change in catalyst loading and amount of product D which can be added to the feed.

4.3.2.Effect of catalyst loading

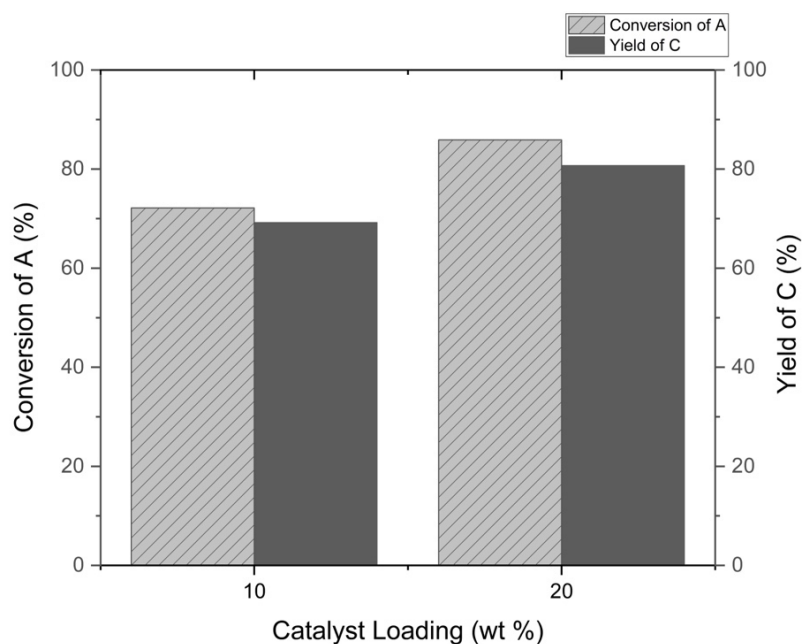


Figure 32 - Effect of Catalyst Loading

As discussed earlier, we examined the effect of catalyst loading on the conversion and yield. It can be seen from Fig. 3 that the conversion of A and yield of C is greater for catalyst loading of 20 % (by wt. of A). The data obtained from the experiments of varying catalyst loading helped us to carry out economic evaluation by taking into consideration the conversion of A, yield of C and material handling loss. The economic evaluation was performed by taking into consideration the industrial scale of production, errors and changes in market costs of the reactants and products. The case with Catalyst Loading of 20 % was

found to be the best case for the reaction of interest. Though the handling loss and material cost for catalyst loading of 20 % is higher, the cost is compensated by the high yield of highly valuable product C. Thus, catalyst loading of 20% (by wt. of A) was selected for the process.

4.3.3. Effect of Feed mole ratio and addition of solvent

As the reaction temperature was increased, the problems like catalyst and oligomers sticking to the walls of the reactor started arising. The catalyst loading of 20 % (by wt. of A) were also fixed. Thus, in order to solve the problem of catalyst and oligomers sticking to the surface we came with a solution of adding an external solvent which could prevent the catalyst from sticking to the surface by keeping it suspended in the product mixture and not allowing it to settle with the oligomers. We tested several solvent systems to solve the problem. However, addition of any external solvent will be resulting in complex and expensive process of separation and purification of Product C from the product mixture. Therefore, we diluted the feed mixture with the co-product D. The major advantage of using co-product D as a component of the feed is that there is no problem of the catalyst sticking to the walls of the reactor with the added ease of separation of the product C from the product mixture.

Subsequently, co-product D was added to the reaction mixture with varying molar ratios. In the first case, there was no presence of D in the feed. In the second set of experiments, the feed mixture was in the molar ratio of A : B : D = 1 : 1.2 : 1 which corresponded to a 20 % by wt. of D. In the third set of experiments, we took the feed mixture in the molar ratio of A : B : D = 1 : 1.2 : 2 corresponding to 40 % by wt. of D.

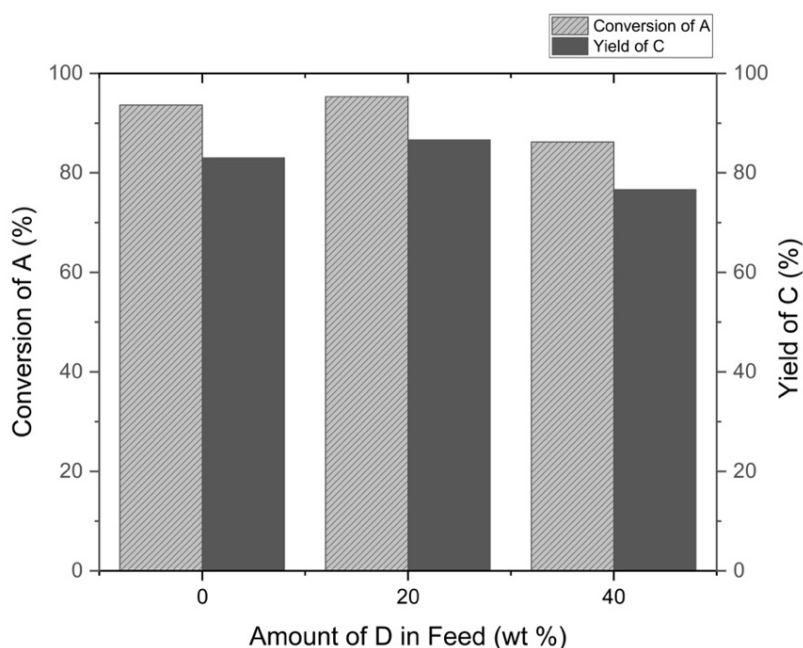


Figure 33 - Effect of amount of D in Feed (wt. %)

It can be seen from Fig. 4. That there is no significant change in the conversion or yield for 0% and 20 % by wt. of D in feed. The 20% by wt. of D in feed greatly prevents the catalyst from sticking to the walls of the reactor thereby allowing ease of operation. At 40 % by wt. of D in the feed, the conversion and the yield of C was seen to decrease significantly. The possible reason for this is the reaction between B and D decreasing the conversion of A and yield of product C.

4.4] Fixed Bed Reactor

By analysing the results of the batch reactor system, we setup a fixed bed reactor in order to carry out the reaction continuously. In order to compare the reactor systems and choose the best possible configuration for carrying out the reaction, we compared the conversion of Reactant A , isolated yields of target product C and co-product D and deactivation of catalyst.

The yields of experiments FBR BBD-7,13,14 were obtained as follows:

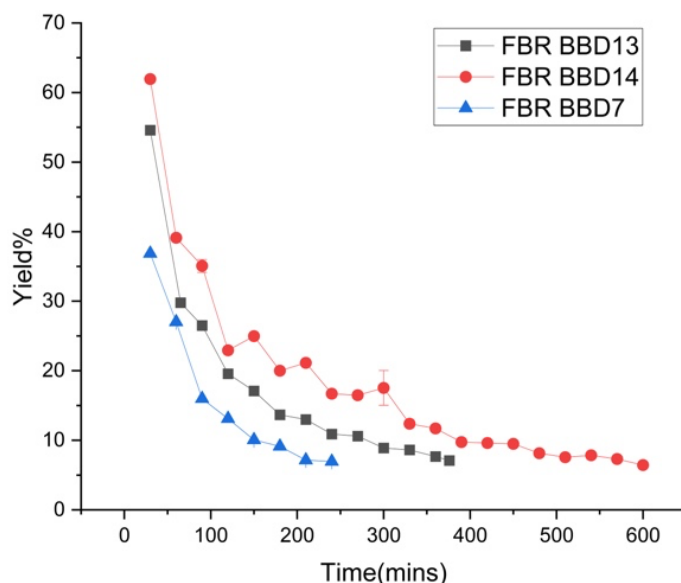


Figure 34 – Yield of C with time

The effect of change in feed flow rate on the yield of the desired product C can be seen clearly through Fig. . In FBR BBD- 7 experiment, the feed flow rate was kept higher at 10 ml/min as compared to FBR BBD-13 and 14 experiments where the feed flow rate was 7 ml/min. It can be seen that at higher flow rates the maximum yield of the product C is significantly lower. The change in flow rates directly affects the residence times of the reactor. Thus, it can be said that at higher flow rates and shorter residence times, the yields are reduced due to insufficient time for mass transfer. The experiments FBR BBD-13 and 14 were repeatability runs taken in order to test the robustness of the system and to replicate the results.

For economical industrial operation, the fixed bed reactor needed to be operated continuously for more than 22 hours without the need of replacing the catalyst. In order to test the relevance and reliability of our fixed bed reactor setup for industrial operations, we carried out continuous reactions on the fixed bed reactor. The reaction conditions were same as FBR BBD 7 and the 22 hour reaction was named FBR BBD 8. At $t = 11$ hours, a stream of fresh D was passed through the fixed bed reactor at 7 ml/min for 30 minutes. During this time, the Feed tank was also refilled so that it could go on for another 11 hours. The conversion of reactant A with respect to time and yield of product C with time are shown in Fig.

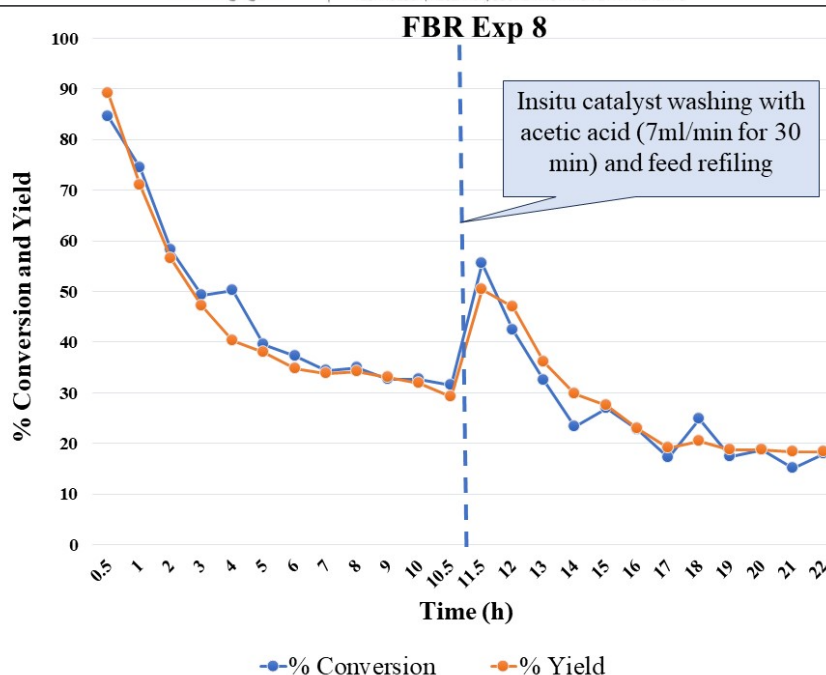


Figure 35 - FBR BBD 8

It can be seen that, at the start of the reaction, the conversion of A is around 85 %. However by the end of the run at $t = 22$ hours, the conversion of A drops to around 18 % indicating severe deactivation of catalyst. After $t = 11$ hours, when D was passed through the fixed bed reactor, sudden improvement in conversion of A and yield of C is seen. The conversion increases from 32 % to 58 % and the yield increases from 28 % to 50 %. This provides promising data to conclude that if D is passed for sufficient time at higher flow rates through the fixed bed reactor, partial regeneration of catalyst is possible. This in situ catalyst washing method was studied extensively in order to design a system where the cycle times between the input and regeneration steps could be fixed for economical operation. The major problem faced during the 22 hour run was that the pressure drop across the fixed bed reactor was seen to increase significantly as the time progressed. The pressure drop at the start of the reaction was ~ 6 bar. At the end of 24 hours, the pressure drop across the column was 90 bar. The large pressure drop was caused primarily because of the oligomerization reactions. The oligomers choked the catalyst causing the pump to increase the pumping power in order to maintain a constant pressure of 6 bar at the outlet. The combined effect of oligomer deposition on the catalyst and large pressure drop caused attrition of catalyst extrudes. The before and after condition of the catalyst can be seen.



Figure 36 - Attrited Catalyst after calcination

It was concluded from the results obtained from the 24 hour long fixed bed reactor run, that the process is not favourable for industrial use.

Batch mode of operation was thus selected in order to produce the product C. The challenges associated with the regeneration of catalyst and material handling were tackled separately in order to deploy the process in the industry.

4.5] Batch Distillation

The target of the industrial process development was to obtain the pure product C. The product mixture containing target product C, Co-product D and trace amounts of unreacted reactants and oligomers was subjected to fractional vacuum distillation to obtain pure C.

The best results were obtained at vacuum of 600 mm of Hg i.e. reduced pressure of 0.2 bar with 99% pure product C and an isolated yield of 54%. In the future course, the distillation setup is to be improved with focus of obtaining a higher isolated yield of product C of more than 80 %.

4.6] Regeneration of catalyst.

Regeneration of catalyst is a crucial point which also affects the type of catalyst to be used. For our reaction, the catalyst is primarily deactivated due to fouling. The large chain oligomers of the reactant and product are deposited on the catalyst thereby blocking the access to active sites for the reactants. The best noted way for regeneration of such deactivated catalyst is by calcining it at 600 °C. The Ion exchange resins have a polymer backbone and thus, regeneration of catalyst by calcining is impossible as it would also

degrade the structure of the ion exchange resin. Zeolites, however, have a very high thermal stability and can withstand temperatures of more than 750 °C.

As zeolites were chosen as the favourable catalyst, a process and equipment for efficient regeneration of the catalyst needed to be designed. The conventional process involves subjecting the deactivated catalyst to extreme temperatures using furnace. However, the furnaces come with inherent disadvantages like low heating efficiency and high energy consumption. In order to eliminate the challenges faced in the conventional method, a design fluidized bed calciner was hypothesized for regenerating the spent catalyst. The investment required for setting up a fluidized bed calciner are comparatively lower. Additionally, fluidized bed calciners can allow large scale continuous regeneration of catalyst.

Fluidized bed dryers are commonly used in the industry for drying the agricultural and chemical products in dispersion or multi-dispersion state. The idea of fluidized bed dryer has been extended for it to operate at higher temperatures which is favourable for regeneration of the catalyst(36).

The solid spent catalyst needs to be completely dispersed in vertically flowing air stream as a consequence of the buoyancy effect of the air. Due to the turbulent activity, good mixing of solid catalyst and high rates of heat transfer between catalyst and air can be obtained. Because of this phenomenon, the catalyst and air temperature can be maintained uniform throughout the fluidized bed calciner. For the fluidized bed calciner to work efficiently, the fluidization of catalyst needs to be effective and for this the gas superficial velocity should be varied between two extreme values. These two extreme values correspond to a minimum fluidization velocity, below which fluidization does not occur and a maximum fluidization velocity, above which entrainment can occur. Thus, for designing the fluidized bed calciner, the minimum and maximum fluidization velocities were calculated.

The minimum fluidization velocity can be given by Equation 1 (37):

$$V_{min} = \left[\frac{4}{3} \frac{dp(\rho_s - \rho_{Ac})}{\rho_{Ac}} \frac{1}{(0.4 + 24/Re + 4/Re^{1/2})} \right]^{1/2} \quad (1)$$

Where,

$$Re = \frac{V_{Ac} \rho_{Ac} dp}{\mu} \quad (2)$$

$$Ar = \frac{(\rho_s - \rho_{Ac}) g \rho_{Ac} dp^3}{\mu^2} \quad (3)$$

The fluidized bed porosity can be expressed as function of Reynolds Number and Archimedes Number using the Todes Equation.

$$\varepsilon = \left(\frac{18 Re + 0.36 Re^2}{Ar} \right)^{0.21} \quad (4)$$

The industrially obtained zeolite catalyst has a diameter (dp) of 80 microns.

$$\therefore dp = 80 \text{ microns} = 80 \times 10^{-6} \text{ metre} \quad (5)$$

The density (ρ_s) of the catalyst was measured to be 1180 kg/m³. The density of fluid i.e. air (ρ_{Ac}) is 1.29 kg/m³. The viscosity of air at temperatures of more than 300 °C was calculated to be 3.178×10^{-5} kg/m s. For the present case as the catalyst to be regenerated is sent to calciner in the form of wet cake, the porosity of the bed can be very low. For the calculations, the porosity of the bed is taken as 0.5, assuming that 50 % of the bed is porous.

Substituting these values in the equation 3, the Archimedes number (Ar) was calculated to be :

$$Ar = 7.554 \quad (6)$$

The value of Archimedes Number and porosity were substituted in the equation 4 and the equation was solved simultaneously to obtain the value of Re .

$$Re = 0.015 \quad (7)$$

Substituting the value of Re in the equation 1, we get :

$$V_{min} = 0.0073 \text{ m/s} \quad (8)$$

Thus, the velocity of 0.0073 m/s is the minimum fluidization velocity required by the catalyst bed of porosity 0.5 to get fluidized. Whenever the air velocity is greater than this value, the catalyst bed will be fluidized.

However, there is also a upper limit to the air velocity. If the value of the air velocity exceeds a critical point, the catalyst particles would no longer be fluidized and would escape or get trapped on the upper wall of the calciner. This is because, the high velocity would cause severe entrainment of the catalyst particles. It is thus necessary to calculate the maximum air velocity in order to design the fluidized bed calciner.

The maximum air velocity can be given by(38,39) :

$$V_{max} = V_{min} \left(11.75 - \frac{0.1046}{1+0.00373 Ar^{0.6}} \right)^1 \quad (9)$$

Substituting the value of V_{min} and A_r in Equation 9, we have

$$V_{max} = 0.08502 \text{ m/s} \quad (10)$$

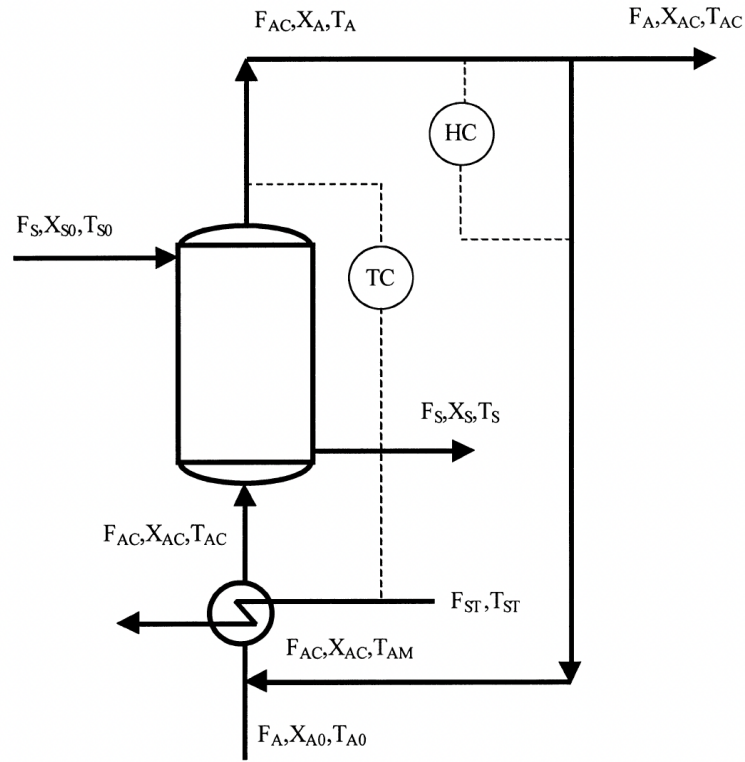


Figure 37 - Fluidized bed Calciner design with controls

5) Future Work

The study of process development for production of solid acid catalysts gave us some interesting insights into catalyst design, reaction kinetics, side reactions including oligomerization, distillation of products and regeneration of catalysts.

The scope of future work primarily involves improvement of the current research and experimental designs for getting better conclusive results. The set of soaking experiments carried out on the Ion Exchange Resin (Amberlyst-15) are to be repeated using a different solid acid catalyst namely zeolite. By varying the Si/Al ratio and analysing the catalyst using ^{31}P NMR, we can obtain valuable insights related to the interaction of reactants and products on acid sites of varying strengths and we can also note the effect of strength and number of acid sites on the oligomerization reactions and thus the deactivation of catalyst. The Thermogravimetric analysis of the spent catalyst obtained from the soaking and deactivation of catalyst experiments (both Ion exchange resin and zeolite) is to be performed. The mass loss with respect to increase in temperature which is recorded in Thermogravimetric analysis can be used to infer the degree of deactivation of catalysts in different conditions.

Sophisticated methods for separation of catalyst from the product mixture which are under investigation should be implemented in the large scale industrial process. The distillation assembly also needs to be worked on. The energy efficiency of the operation should be increased before scaling up the assembly to distil large quantities of product mixtures. The economic evaluation of the distillation operation also needs to be carried out to compare vacuum distillation with the normal distillation operation. Based on the theoretical design, a small scale model of the fluidized bed calciner for regeneration of catalyst can also be developed.

6) Conclusions

A sustainable, efficient and cost competitive process for production of an API Precursor “C” was developed. The developed process was tested rigorously for validating the safety, reliability and consistency. Future scope of work was defined to improve some of the process parameters and also help during the scale up operation.

References :

1. Solid acid catalysts Avelino Corma. 1359;
2. Primo A, Garcia H. Zeolites as catalysts in oil refining. *Chem Soc Rev.* 2014 Mar 27;43(22):7548–61.
3. Cinar S, Beler-Baykal B. Ion exchange with natural zeolites: an alternative for water softening? *Water Science and Technology.* 2005 Jun 1;51(11):71–7.
4. Palčić A, Valtchev V. Analysis and control of acid sites in zeolites. Vol. 606, *Applied Catalysis A: General.* Elsevier B.V.; 2020.
5. Vijay S. ION EXCHANGE RESINS AND THEIR APPLICATIONS. *Journal of Drug Delivery & Therapeutics* [Internet]. 2011;2014(4):115–23. Available from: <http://jddtonline.info>
6. Barbara P, Liguori F. Ion exchange resins: Catalyst recovery and recycle. Vol. 109, *Chemical Reviews.* 2009. p. 515–29.
7. Corma A. Inorganic Solid Acids and Their Use in Acid-Catalyzed Hydrocarbon Reactions. *Chem Rev.* 1995 May 1;95(3):559–614.
8. Niwa M, Suzuki K, Katada N, Kanougi T, Atoguchi T. Ammonia IRMS-TPD Study on the Distribution of Acid Sites in Mordenite. *J Phys Chem B.* 2005 Oct 1;109(40):18749–57.
9. Zecchina A, Spoto G, Bordiga S. Probing the acid sites in confined spaces of microporous materials by vibrational spectroscopy. *Physical Chemistry Chemical Physics.* 2005;7(8):1627.
10. Ping Z. In situ FTIR–attenuated total reflection spectroscopic investigations on the base–acid transitions of polyaniline. Base–acid transition in the emeraldine form of polyaniline. *J Chem Soc, Faraday Trans.* 1996;92(17):3063–7.
11. Auroux A. Acidity characterization by microcalorimetry and relationship with reactivity. *Top Catal.* 1997;4(1/2):71–89.
12. Cardona-Martinez N, Dumesic JA. Applications of Adsorption Microcalorimetry to the Study of Heterogeneous Catalysis. In 1992. p. 149–244.
13. Benesi HA. Acidity of Catalyst Surfaces. I. Acid Strength from Colors of Adsorbed Indicators. *J Am Chem Soc.* 1956 Nov 1;78(21):5490–4.
14. BAKSHI K. Characterization of acid-base catalysts by calorimetric titration I. Correlation with alcohol dehydration activity. *J Catal.* 1975 Jun;38(1–3):312–25.

15. Sulikowski B, Datka J, Gil B, Ptaszynski J, Klinowski J. Acidity and Catalytic Properties of Realuminated Zeolite Y. *J Phys Chem B*. 1997 Aug 1;101(35):6929–32.
16. Shah DR, Nezam I, Zhou W, Proaño L, Jones CW. Isomorphous Substitution in ZSM-5 in Tandem Methanol/Zeolite Catalysts for the Hydrogenation of CO₂ to Aromatics. *Energy & Fuels*. 2024 Feb 1;38(3):2224–34.
17. Hoffman RE. Standardization of chemical shifts of TMS and solvent signals in NMR solvents. *Magnetic Resonance in Chemistry*. 2006 May 14;44(6):606–16.
18. Balci M. Dynamic NMR Spectroscopy. In: *Basic ¹H- and ¹³C-NMR Spectroscopy*. Elsevier; 2005. p. 213–31.
19. HUNGER M. Brønsted Acid Sites in Zeolites Characterized by Multinuclear Solid-State NMR Spectroscopy. *Catalysis Reviews*. 1997 Nov;39(4):345–93.
20. Köhl O. The Range of Chemical Shifts, Coupling Constants, and What Influences Each. In: *Phosphorus-31 NMR Spectroscopy*. Berlin, Heidelberg: Springer Berlin Heidelberg; p. 7–23.
21. Zheng A, Huang SJ, Liu S Bin, Deng F. Acid properties of solid acid catalysts characterized by solid-state ³¹P NMR of adsorbed phosphorous probe molecules. *Physical Chemistry Chemical Physics*. 2011 Sep 7;13(33):14889–901.
22. Lunsford JH, Sang H, Campbell SM, Liang CH, Anthony RG. An NMR study of acid sites on sulfated-zirconia catalysts using trimethylphosphine as a probe. *Catal Letters*. 1994;27(3–4):305–14.
23. Zasukhin DS, Kasyanov IA, Kolyagin YG, Bulygina AI, Kharas KC, Ivanova II. Evaluation of Zeolite Acidity by ³¹P MAS NMR Spectroscopy of Adsorbed Phosphine Oxides: Quantitative or Not? *ACS Omega*. 2022 Apr 12;7(14):12318–28.
24. Hayashi S. Adsorption of Trimethylphosphine Oxide Molecules from the Gas Phase to Probe Surface Acidity by Solid-state NMR. *Chem Lett*. 2009 Oct 5;38(10):960–1.
25. Chronister CW, Drago RS. Determination of hydrogen-bonding acid sites on silica using the Cal-Ad method. *J Am Chem Soc*. 1993 Jun 1;115(11):4793–8.
26. Kao HM, Chang PC, Liao YW, Lee LP, Chien CH. Solid-state NMR characterization of the acid sites in cubic mesoporous Al-MCM-48 materials using trimethylphosphine oxide as a ³¹P NMR probe. *Microporous and Mesoporous Materials*. 2008 Sep;114(1–3):352–64.

27. Zhao Q, Chen WH, Huang SJ, Liu SB. 39 Qualitative and quantitative determination of acid sites on solid acid catalysts. In 2003. p. 205–9.
28. Yi X, Zheng A, Liu SB. Acidity characterization of solid acid catalysts by solid-state ^{31}P NMR of adsorbed phosphorus-containing probe molecules: An update. In 2020. p. 65–149.
29. Xu J, Wang Q, Li S, Deng F. Solid-State NMR Characterization of Acid Properties of Zeolites and Solid Acid Catalysts. In 2019. p. 159–97.
30. Pires E, Fraile JM. Study of interactions between Brønsted acids and triethylphosphine oxide in solution by ^{31}P NMR: evidence for 2 : 1 species. *Physical Chemistry Chemical Physics*. 2020;22(42):24351–8.
31. Lanza fame P, Papanikolaou G, Perathoner S, Centi G, Giordano G, Migliori M. Weakly acidic zeolites: A review on uses and relationship between nature of the active sites and catalytic behaviour. *Microporous and Mesoporous Materials*. 2020 Jun 15;300.
32. Sinha AK, Hegde SG, Jacob NE, Sivasanker S. Preparation and characterization of the silicoaluminophosphate SAPO-39. *Zeolites*. 1997 May;18(5–6):350–5.
33. Yi X, Zheng A, Liu SB. Acidity characterization of solid acid catalysts by solid-state ^{31}P NMR of adsorbed phosphorus-containing probe molecules: An update. In 2020. p. 65–149.
34. Zheng A, Huang SJ, Liu SB, Deng F. Acid properties of solid acid catalysts characterized by solid-state ^{31}P NMR of adsorbed phosphorous probe molecules. *Physical Chemistry Chemical Physics*. 2011;13(33):14889.
35. Artigues A, Puy N, Bartrolí J, Fábregas E. Comparative Assessment of Internal Standards for Quantitative Analysis of Bio-oil Compounds by Gas Chromatography/Mass Spectrometry Using Statistical Criteria. *Energy & Fuels*. 2014 Jun 19;28(6):3908–15.
36. Kiranoudis CT, Maroulis ZB, Marinou-Kouris D. Design and operation of convective industrial dryers. *AIChE Journal*. 1996 Nov 17;42(11):3030–40.
37. Krokida MK, Kiranoudis CT. Pareto design of fluidized bed dryers. Vol. 79, *Chemical Engineering Journal*. 2000.
38. Kokaulin VD, Sorokin AA. Classification of methods for drying dispersed materials in a suspended state. *Glass and Ceramics*. 1983 Aug;40(8):414–5.
39. Krokida MK, Kiranoudis CT. Pareto design of fluidized bed dryers. Vol. 79, *Chemical Engineering Journal*. 2000.